Lab Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Understanding	3					
Analysis	4					
Implementation	8					
Report Writing	10					
					Total obtained mark	
Comments						

Semester: Fall 2025

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Batch: 46 **Section: B**

Course Code: SE112 **Course Name: Computer Fundamentals Lab**

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Submission Date: 14 /12/2025

Microsoft Word

Microsoft Word is one of the most popular word processing programs. It is flexible and easy to use for creating, editing, and formatting documents. Whether you're writing a simple letter, preparing a resume, or working on a formal report, MS Word offers a great set of tools to help you create professional and well-structured content. With features like text formatting, inserting tables and images, page layout options, and built-in templates, users can manage a variety of document tasks efficiently.

Follow these simple steps to open MS Word on your personal computer:

Start* → *All Programs* → *MS Office* → *MS Word.

Microsoft Word Interface

The Microsoft Word interface is designed for creating and editing documents.

File Menu: The File tab will bring you into the Backstage View. The Backstage View is where you manage your files and the data about them – creating, opening, printing, saving, inspecting for hidden metadata or personal information, and setting options.

Ribbon: An area across the top of the screen that makes almost all the capabilities of Word available in a single area.

Tabs: An area on the Ribbon that contains buttons that are organized in groups. The default tabs are Home, Insert, Design, Layout, References, Mailings, Review, View and EndNote X5.

Title Bar: A horizontal bar at the top of an active document. This bar displays the name of the document and application. At the right end of the Title Bar is the Minimize, Maximize and Close buttons.

Groups: A group of buttons on a tab that are exposed and easily accessible.

Dialog Box Launcher: A button in the corner of a group that launches a dialog box containing all the options within that group.

Status Bar: A horizontal bar at the bottom of an active window that gives details about the document.

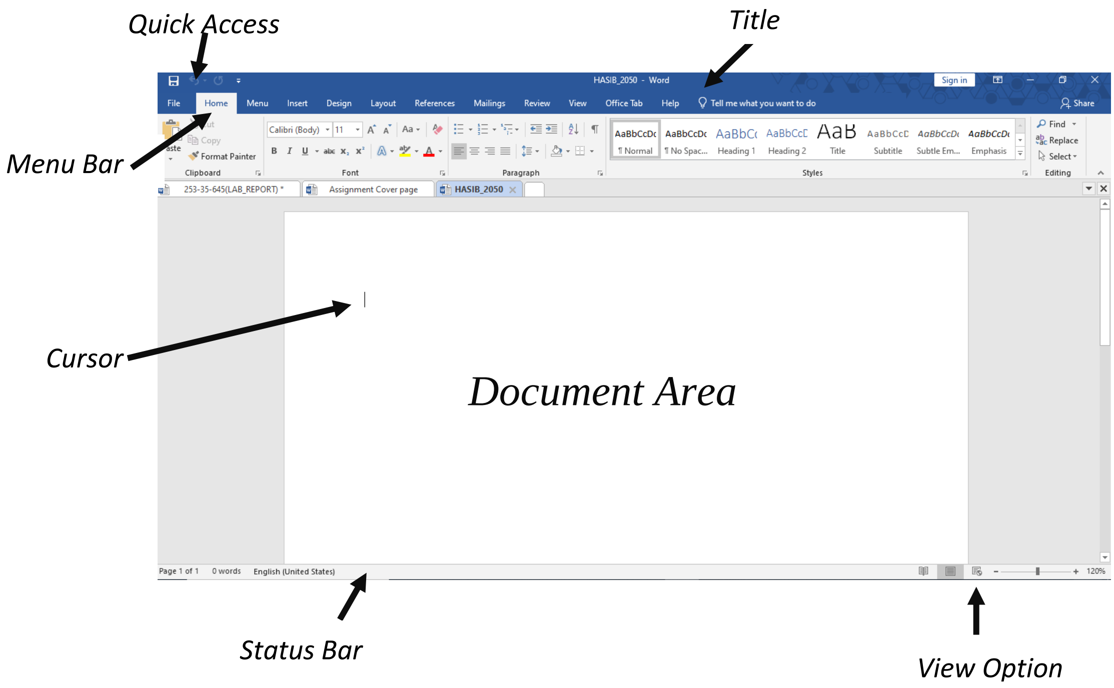
View Toolbar: A toolbar that enables, adjusts, and displays different views of a document.

Zoom: Magnifies or reduces the contents in the document window.

Quick Access Toolbar: A customizable toolbar at the top of an active document. By default, the Quick Access Toolbar displays the Save, Undo, and Repeat buttons and is used for easy access to frequently used commands. To customize this toolbar, click on the dropdown arrow and select the commands you want to add.

Tell Me: This is a text field where you can enter words and phrases about what you want to do next and quickly get to features you want to use or actions you want to perform. You can also use Tell Me to find help about what you're looking for, or to use Smart Lookup to research or define the term you entered.

Contextual Tabs are designed to appear on the Ribbon when certain objects or commands are selected. These tabs provide easy access to options specific to the selected object or command. For example, the commands for editing a picture will not be available until the picture is selected, at which time the **Picture Tools** tab will appear. I will discuss it the later part in **Drawing Tools** and **Picture Tools** .

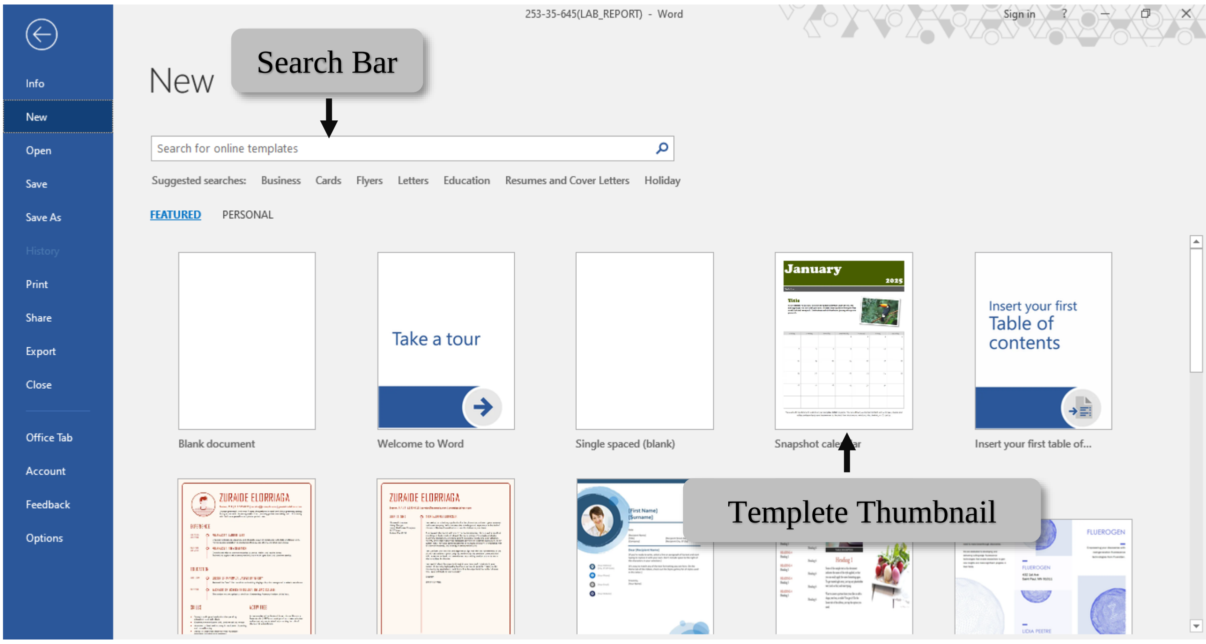


Microsoft Word Tabs Explained

Now let us discuss the tabs and components of MS Word. With these tabs, you can perform different operations on your documents. You can create, delete, style, modify, or view the content of your document.

1.FILE

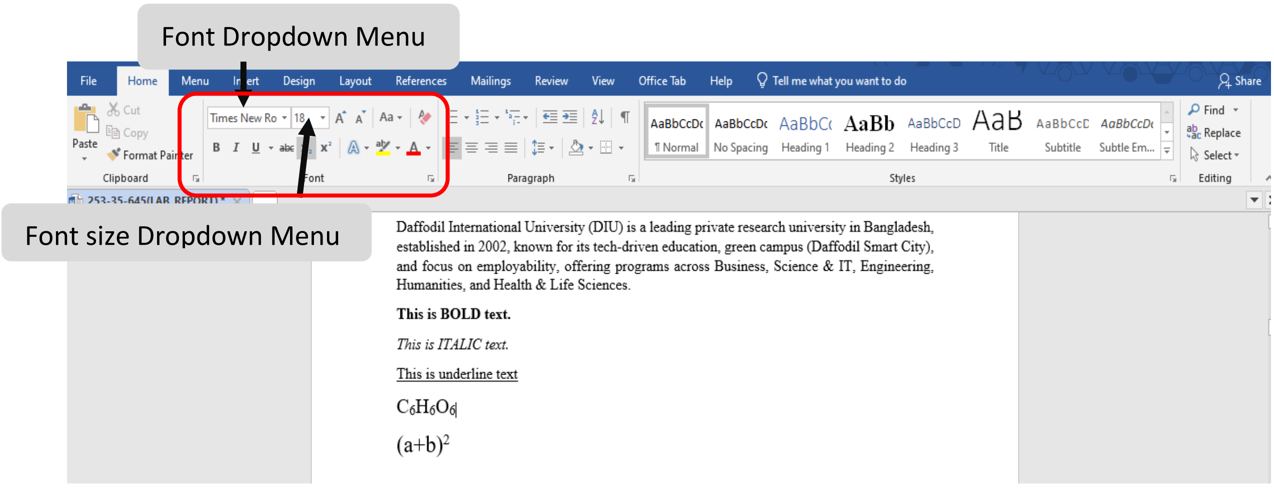
It includes options related to the file, such as New (to create a new document), Open (to open an existing document), Save (to save the document), Save As (to save documents), History, Print, Share, Export, and Info.

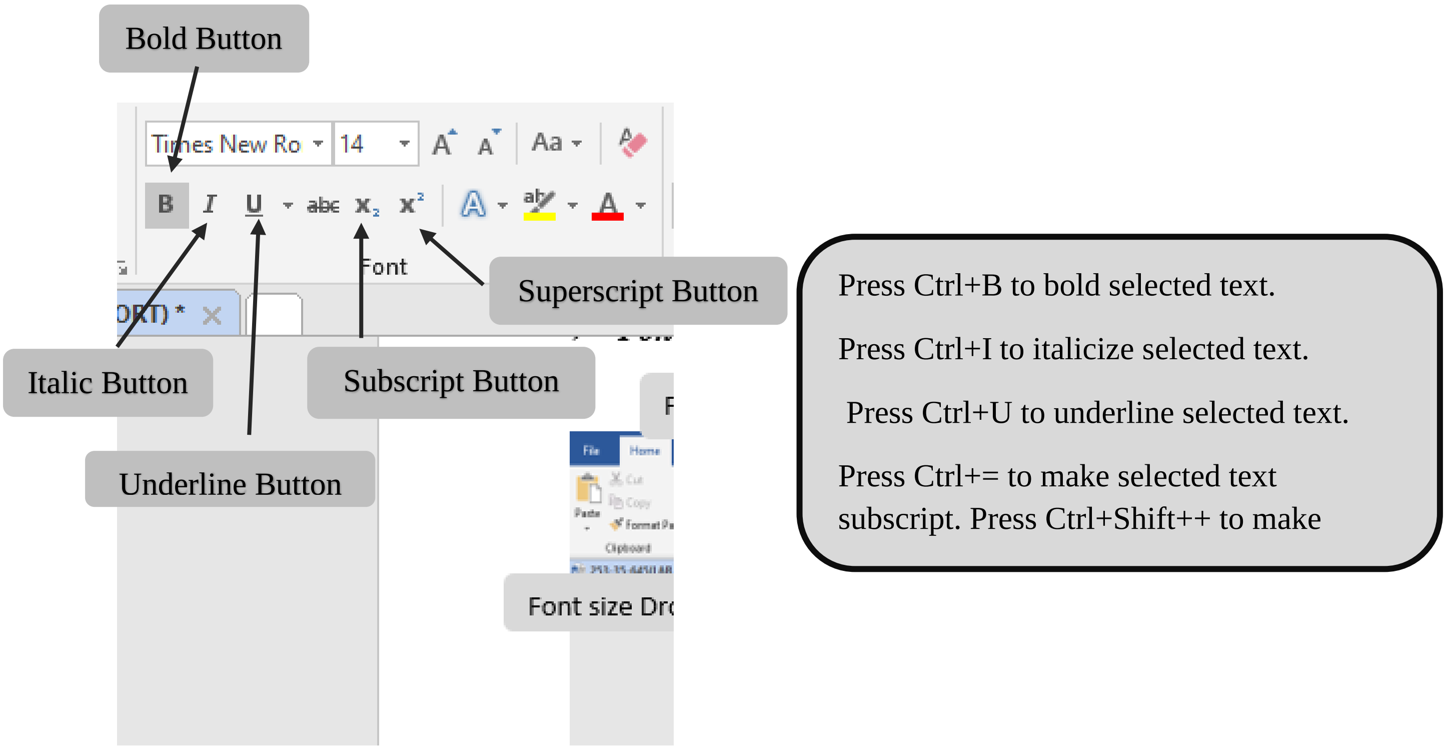


2. Home

It is the default tab of MS Word and it is generally divided into five groups, i.e., Clipboard, Font, Paragraph, Style and Editing. It allows you to select the color, font, emphasis, bullets, position of your text. It also contains options like cut, copy, and paste. After selecting the home tab, you will get below options:

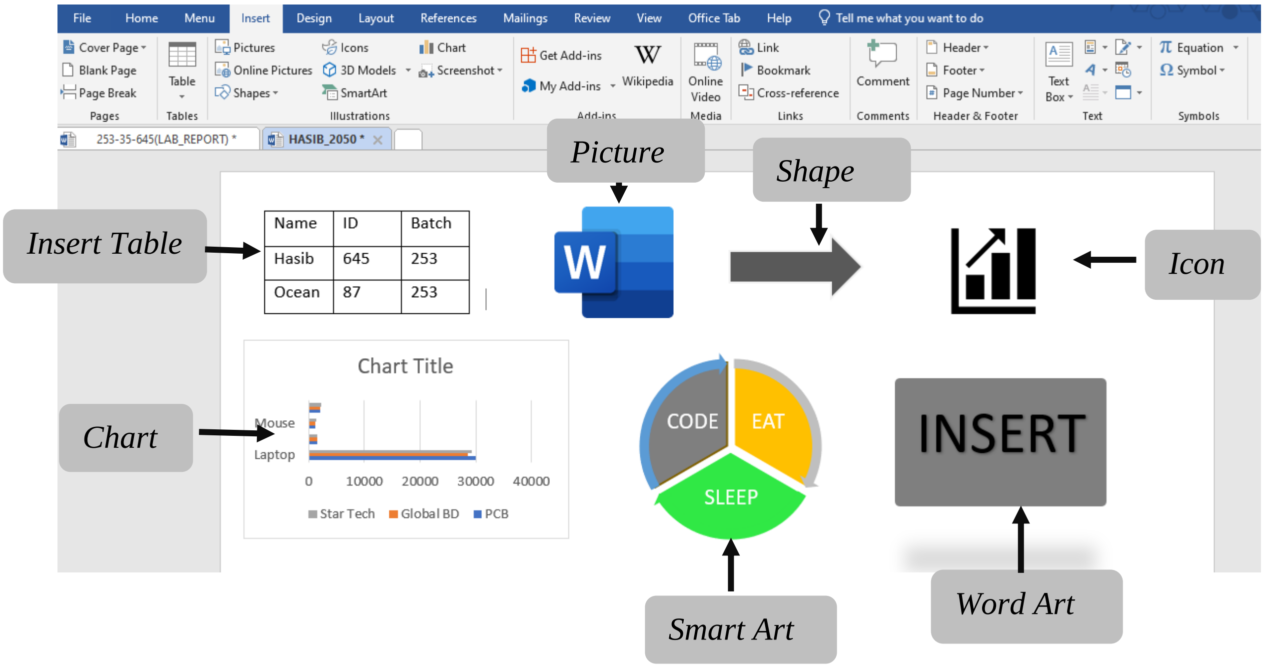
➢ ***Font***





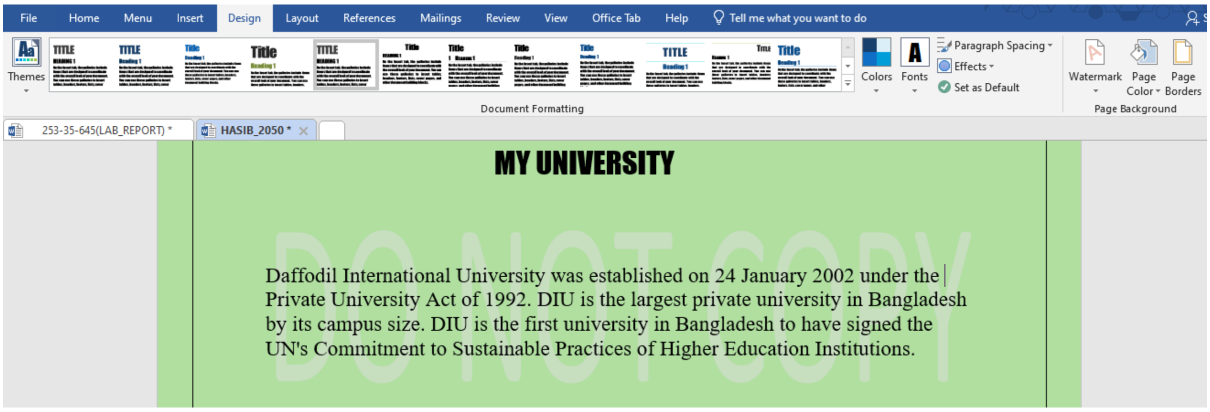
3. Insert

Add content like tables, images, hyperlinks, charts, word art, date and time, headers, footers, shapes, text boxes, equations, and more to your document.



4. Design

It is the fourth tab in the ribbon or menu bar. You can choose from a variety of document designs on the design tab, including ones with centered titles, offset headings, left-justified text, page borders, watermarks, page color, and more, as seen in the image below:



5.Layout

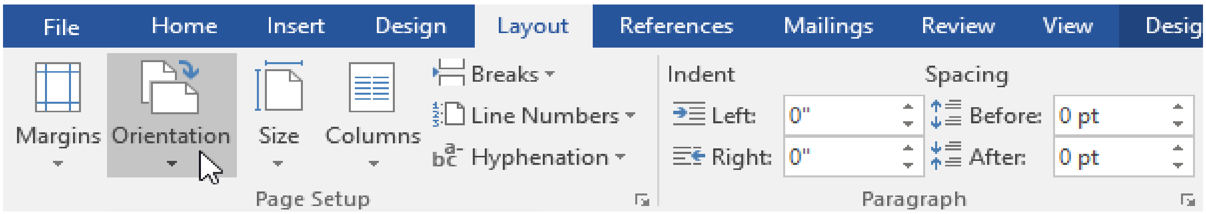
On the menu bar or ribbon, it is the fifth tab. It contains all the settings that let you customize the layout of the pages in your Microsoft Word document. As seen in the image below, it has features like line breaks, control page orientation and size, set margins, display line numbers, set paragraph indentation, and lines apply themes:

- **Page orientation:**

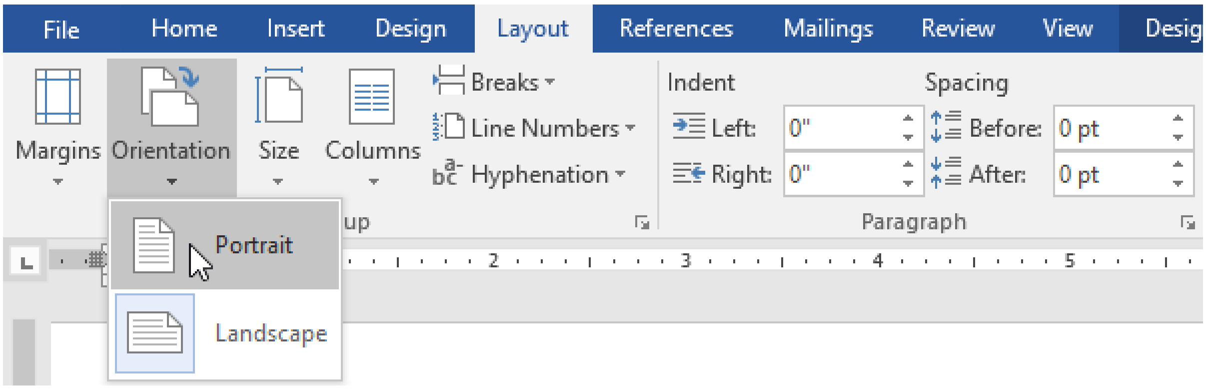
Word offers two page orientation options: landscape and portrait. Landscape means the page is oriented horizontally. Portrait means the page is oriented vertically.

***To change page orientation* :**

1. Select the Layout tab.
2. Click the Orientation command in the Page Setup group.



3. A drop-down menu will appear. Click either Portrait or Landscape to change the page orientation.



4. The page orientation of the document will be changed.

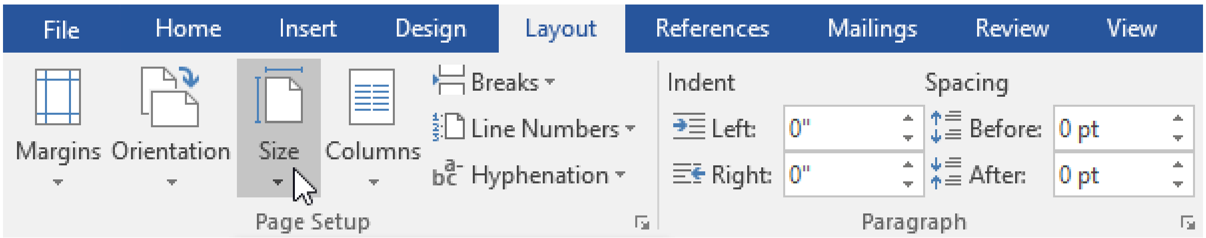
➢ Page size

By default, the **page size** of a new document is 8.5 inches by 11 inches. Depending on your project, you may need to adjust your document's page size. It's important to note that before modifying the default page size, you should check to see which page sizes your printer can accommodate.

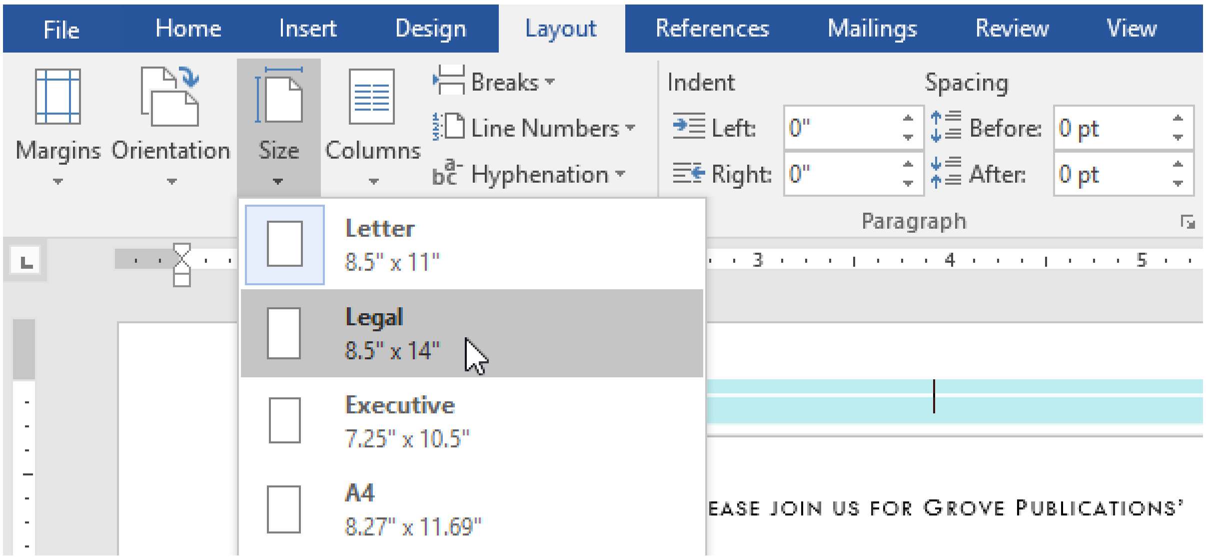
To change the page size:

Word has a variety of **predefined page sizes** to choose from.

1. Select the **Layout** tab, then click the **Size** command.



2. A drop-down menu will appear. The current page size is highlighted. Click the desired **predefined page size**.



3. The page size of the document will be changed.

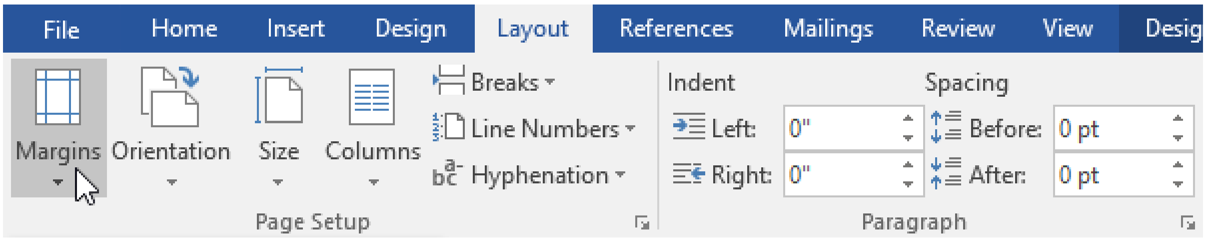
➢ Page margins

A **margin** is the **space** between the text and the edge of your document. By default, a new document's margins are set to **Normal**, which means it has a one-inch space between the text and each edge. Depending on your needs, Word allows you to change your document's margin size.

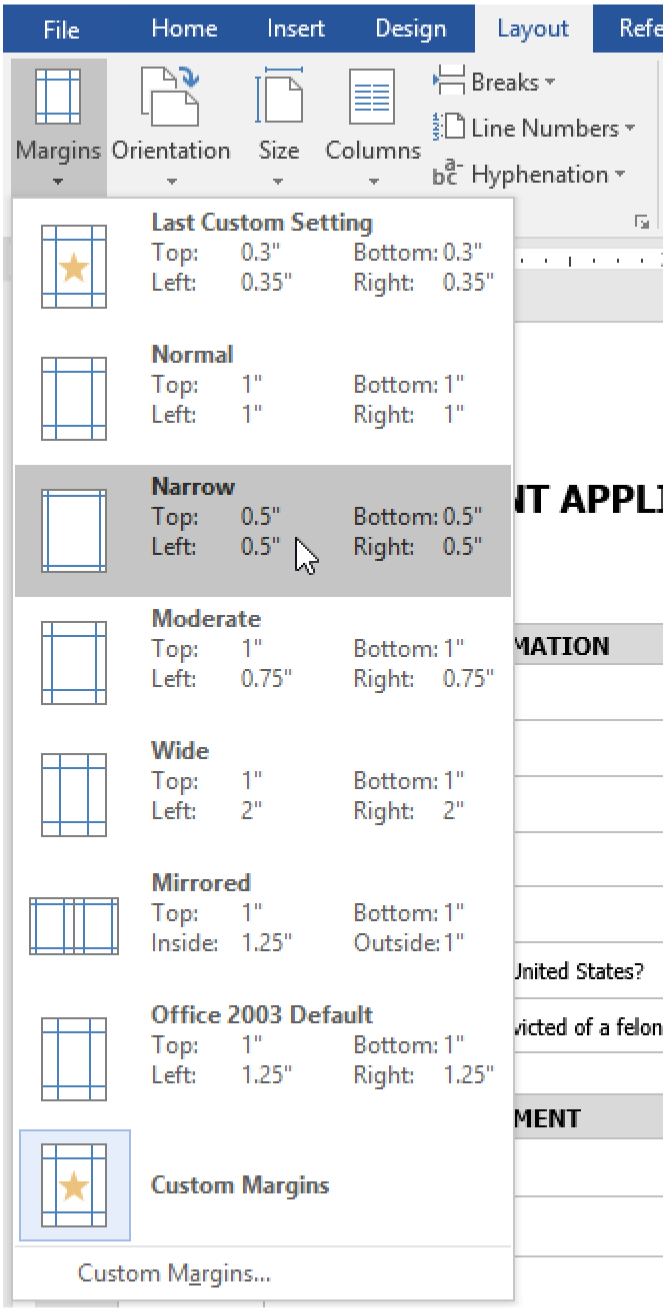
To format page margins:

Word has a variety of **predefined margin sizes** to choose from.

1. Select the **Layout** tab, then click the **Margins** command.



2. A drop-down menu will appear. Click the **predefined margin size** you want.

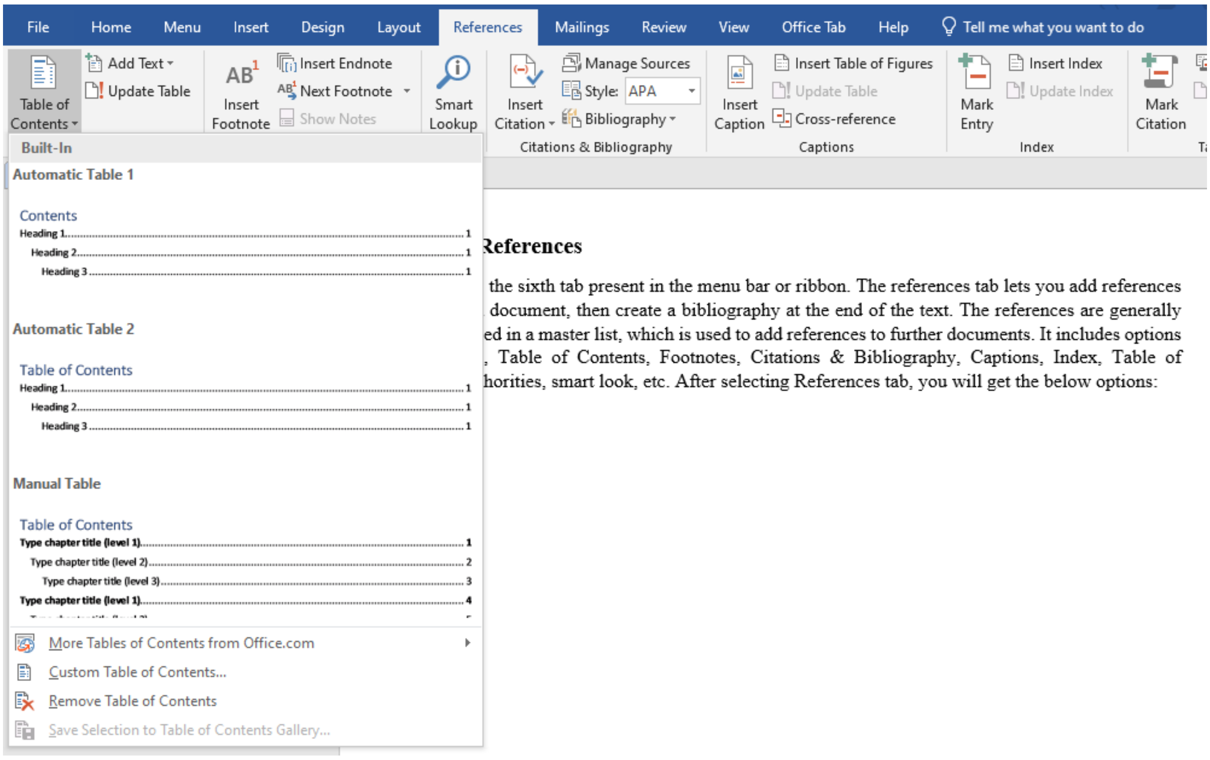


- ***Last Custom Setting*** – Shows the most recent margin values that the user manually set.
- ***Normal*** – Sets standard margins (1 inch on all sides). Used for most documents.
- ***Narrow*** – Reduces margins (0.5 inch on all sides) to fit more content on a page.
- ***Moderate*** – Uses slightly smaller side margins than Normal for a balanced layout.
- ***Wide*** – Increases left and right margins, often used for comments or binding.
- ***Mirrored*** – Sets inside and outside margins for double-sided printing (books or reports).
- ***Office 2003 Default*** – Applies the margin settings used in older versions of Word.
- ***Custom Margins*** – Opens a dialog box where users can set exact margin values manually.

3. The margins of the document will be changed.

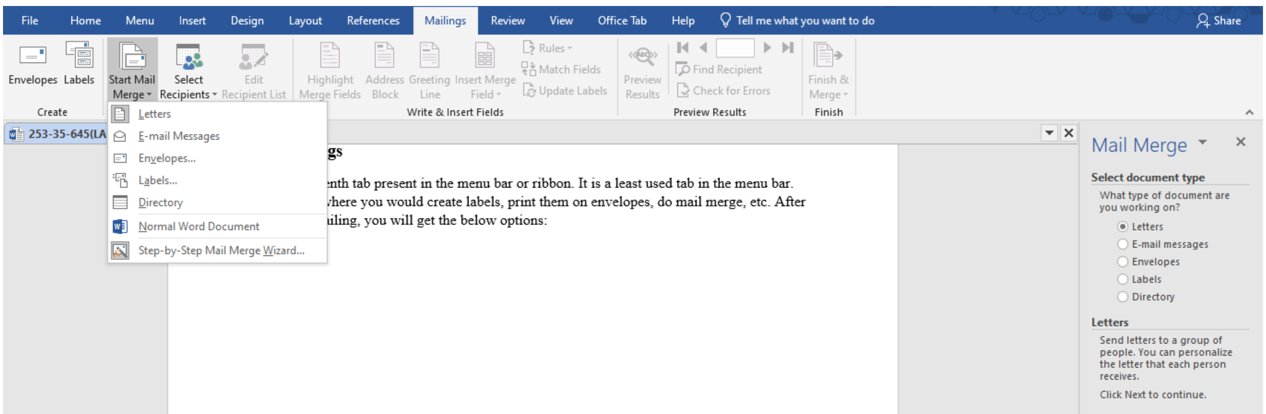
6. References

It is the sixth tab present in the menu bar or ribbon. The references tab lets you add references to a document, then create a bibliography at the end of the text. The references are generally stored in a master list, which is used to add references to further documents. It includes options like, Table of Contents, Footnotes, Citations & Bibliography, Captions, Index, Table of Authorities, smart look, etc. After selecting References tab, you will get the below options:



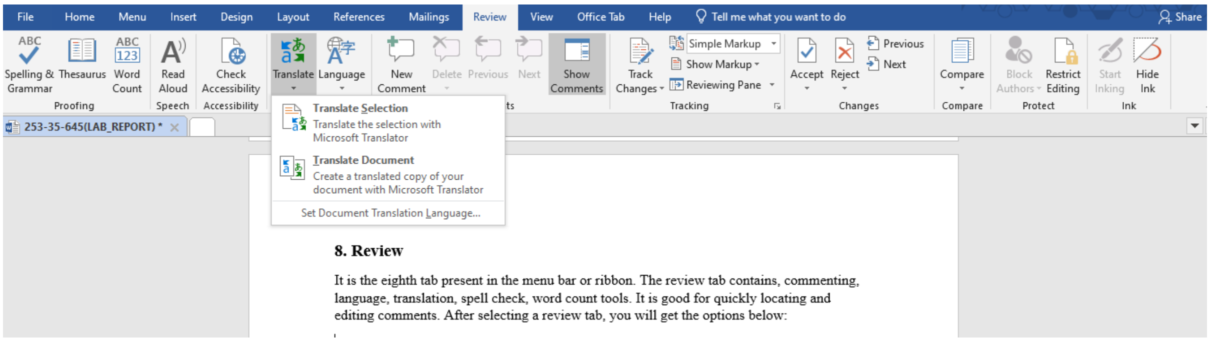
7. Mailings

It is the seventh tab present in the menu bar or ribbon. It is a least used tab in the menu bar. This tab is where you would create labels, print them on envelopes, do mail merge, etc. After selecting mailing, you will get the below options:



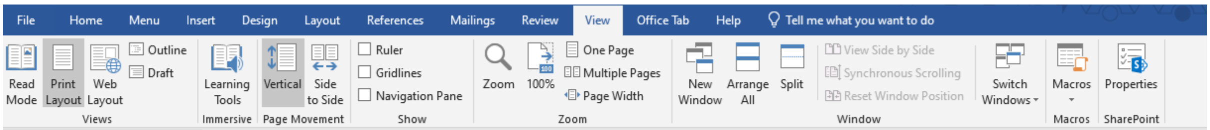
8. Review

It is the eighth tab present in the menu bar or ribbon. The review tab contains, commenting, language, translation, spell check, word count tools. It is good for quickly locating and editing comments. After selecting a review tab, you will get the options below:



10. View

It is the ninth tab present in the menu bar or ribbon. View tab allows you to switch between single page or double page and also allows you to control the layout tools It includes print layout, outline, web layout, task pane, toolbars, ruler, header and footer, footnotes, full-screen view, zoom, etc. as shown in the below image:



11. Formatting Tools for Pictures and Drawings in Microsoft Word

Microsoft Word provides a powerful set of formatting tools for pictures and drawings that help users enhance the visual appearance of documents. These tools allow users to adjust image quality, apply styles and effects, control text wrapping, and properly align visual elements with text. By using picture and drawing format options, documents become more attractive, clear, and professionally organized. These features are especially useful for creating assignments, reports, posters, and presentations where visual content plays an important role.

A. Format Tools for Pictures

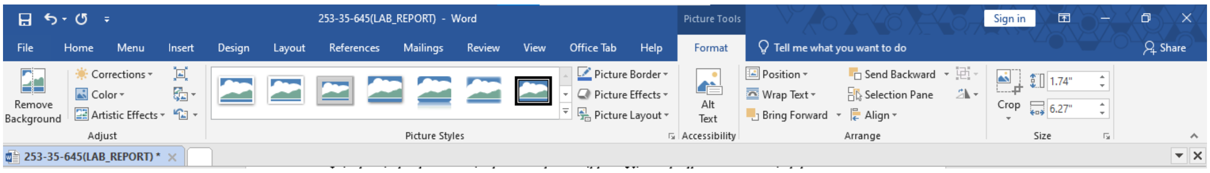
The Picture Format Tab in Microsoft Word appears automatically when a picture is selected. It contains a variety of tools that allow users to modify, enhance, and arrange images within a document. Using these tools, users can adjust image appearance, apply styles and effects, control how text wraps around pictures, and resize or crop images. The Picture Format Tab helps make documents more visually appealing, clear, and professionally presented.

➢ **Adjust Group**

- Remove Background – Removes unwanted background areas from an image.
- Corrections – Adjusts brightness, contrast, and sharpness.
- Color – Changes image color tone, saturation, or converts to grayscale.
- Artistic Effects – Applies visual effects like blur, sketch, or paint style.
- Transparency – Makes the picture partially transparent.

➢ Picture Styles Group

- Picture Styles – Applies pre-designed frames, borders, and shadows.
- Picture Border – Changes border color, thickness, and style.
- Picture Effects – Adds effects such as shadow, reflection, glow, and 3D rotation.



➢ Arrange Group

- Position – Places the picture at a fixed location on the page.
- Wrap Text – Controls how text flows around the picture (Square, Tight, Behind Text, etc.).
- Bring Forward / Send Backward – Changes the layer order of images.
- Align – Aligns pictures with margins or other objects.
- Group / Ungroup – Combines multiple objects into one.
- Rotate – Rotates or flips the picture.

➢ Size Group

- Height & Width – Resizes the picture precisely.
- Crop – Removes unwanted parts of the image.
- Aspect Ratio Lock – Keeps image proportion while resizing.

B. Format Tools for Drawings / Shapes

➢ Insert Shapes

- Edit Shape – Changes the shape type or edits its points.
- Add Text – Inserts text inside the shape.
- Shape Styles Group
- Shape Fill – Fills the shape with color, gradient, picture, or texture.
- Shape Outline – Changes border color, weight, and line style.
- Shape Effects – Adds shadow, glow, reflection, bevel, and 3D effects.

➢ WordArt Styles

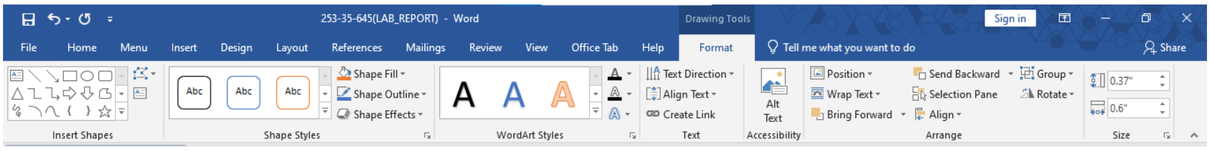
- Text Fill – Changes text color inside the shape.
- Text Outline – Adds or edits text borders.
- Text Effects – Applies effects like shadow, glow, and transform.

➢ Arrange Group

- Position, Wrap Text, Align, Group, Rotate tools work similarly for drawings.

➢ Size Group

- Adjusts height, width, and scaling of shapes.



12. Conclusion

In conclusion, this lab report provided a detailed overview of the major tools and features of Microsoft Word along with their visual representations. Through this practical exploration, we gained a clear understanding of how different components such as menus, ribbons, and formatting tools work together to create, edit, and manage documents efficiently. This lab enhanced our practical skills and improved our confidence in using Microsoft Word for academic and professional purposes. Overall, the knowledge gained from this lab will be helpful for preparing well-structured documents in future tasks.

Microsoft PowerPoint

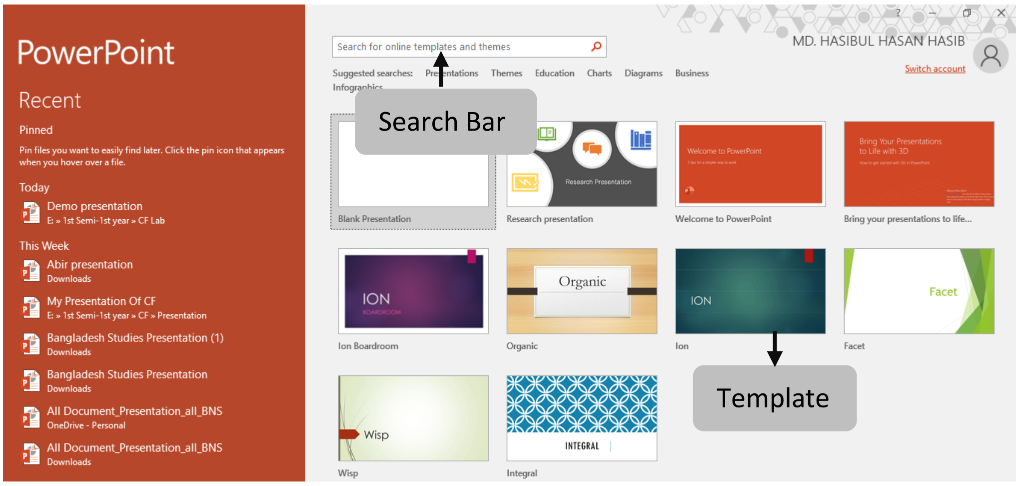
Microsoft PowerPoint is a popular presentation software developed by Microsoft and included in the Microsoft Office suite. It is widely used in education, business, training sessions, seminars, and professional environments to present information in a clear, organized, and visually attractive way. PowerPoint allows users to create slide-based presentations that help communicate ideas effectively to an audience.

Using Microsoft PowerPoint, users can insert and format text, images, charts, tables, shapes, SmartArt graphics, and multimedia elements such as audio and video. The software also provides various themes, slide layouts, and design tools that help maintain consistency and improve the overall appearance of a presentation. Features like animations and slide transitions make presentations more engaging and help emphasize important points.

Microsoft PowerPoint also supports easy editing, formatting, and integration with other Microsoft Office applications such as Word and Excel. This makes it convenient to prepare presentations using data, reports, and documents from different sources. Overall, Microsoft PowerPoint is an essential tool for creating professional, informative, and visually appealing presentations, making it suitable for practical work and lab report documentation.

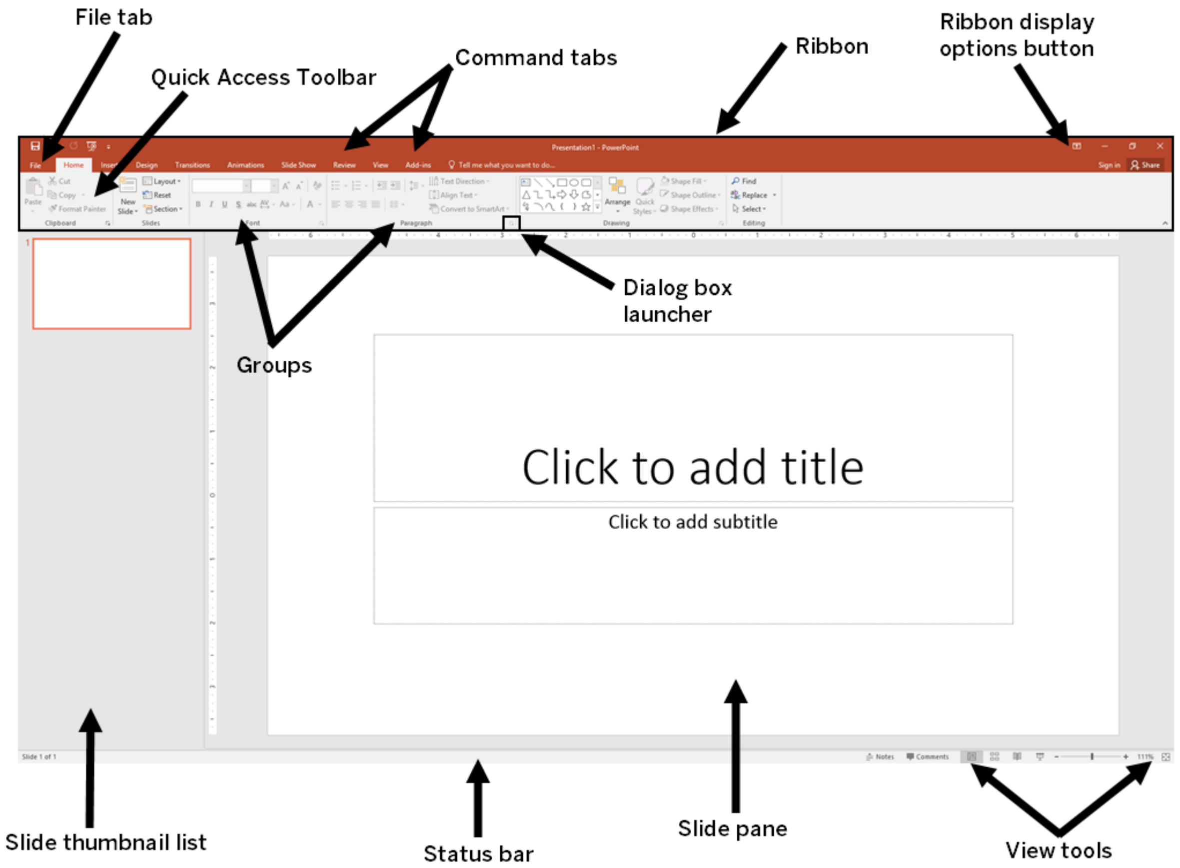
1. PowerPoint Start Screen

PowerPoint Start Screen is the first screen that appears when Microsoft PowerPoint is opened. It allows users to quickly create a new presentation by choosing a blank presentation or selecting from available templates and themes. The Start Screen also provides access to recently opened files, making it easy to continue working on previous presentations.



2.Powerpoint Interface

Microsoft Office Interface refers to the overall layout and visual structure of Microsoft Office applications that helps users interact with the software easily and efficiently. It includes key elements such as the title bar, ribbon, tabs, groups, quick access toolbar, and status bar. This interface is designed to organize tools and commands logically, allowing users to access features quickly, perform tasks smoothly, and work productively across different Microsoft Office programs.

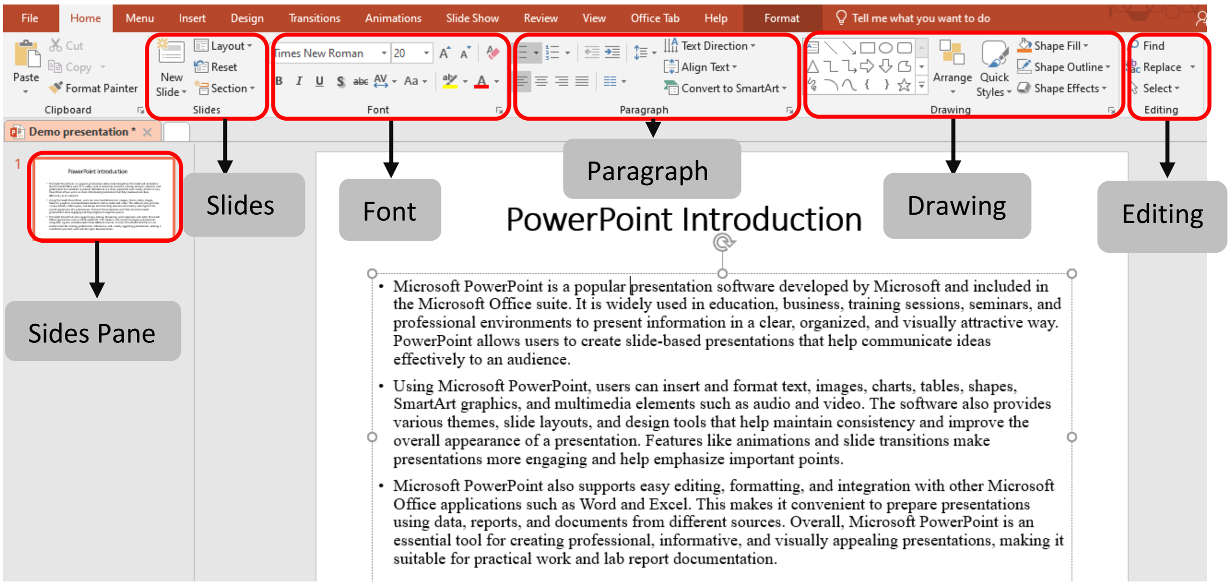


- **File tab:** Opens Backstage view to manage files such as New, Open, Save, Print, and Options.
- **Quick Access Toolbar:** Provides quick access to frequently used commands like Save, Undo, and Redo.
- **Command tabs:** Tabs (Home, Insert, Design, etc.) that organize related tools and commands.
- **Ribbon:** The main toolbar area that displays commands related to the selected tab.
- **Ribbon display options button:** Controls how the Ribbon is shown (auto-hide, show tabs, or show full ribbon).
- **Groups:** Sections within a Ribbon tab that organize related commands (e.g., Font, Paragraph).
- **Dialog box launcher:** A small arrow that opens a dialog box with more detailed settings for a group.

- **Slide thumbnail list:** Displays thumbnails of all slides, allowing quick navigation and reordering.
- **Status bar:** Shows presentation information such as slide number, notes, and view status.
- **Slide pane:** The main working area where the current slide content is created and edited.
- **View tools:** Buttons to change slide views (Normal, Slide Sorter, Reading, Slide Show).

3.Home

MS PowerPoint Home Tab is the main working tab in Microsoft PowerPoint that contains the most commonly used tools for creating and editing slides. It includes options for adding and formatting text, inserting new slides, changing slide layouts, adjusting fonts and paragraphs, and managing objects such as shapes and images. The Home tab helps users perform basic editing tasks quickly and efficiently while designing presentations.



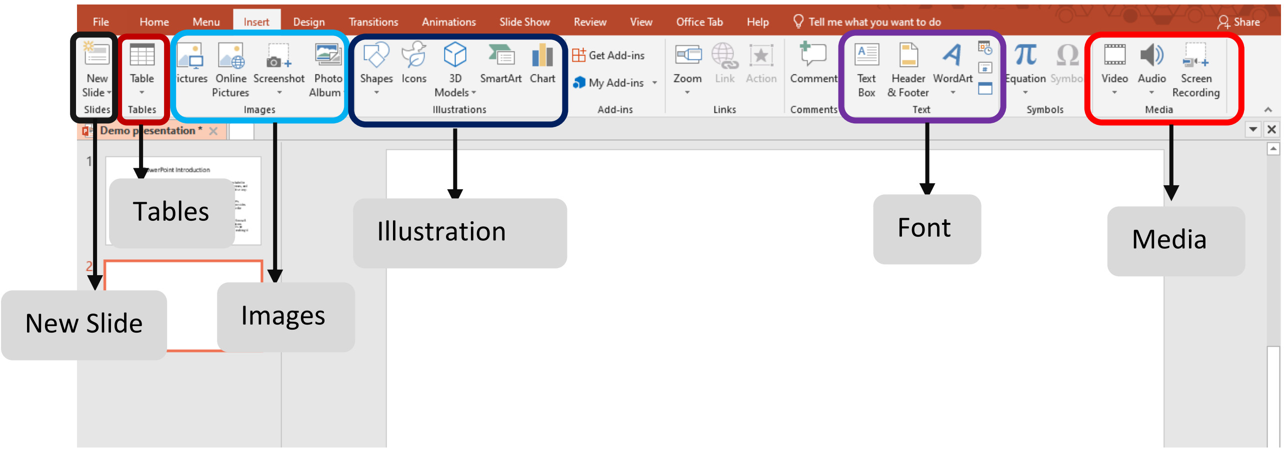
Home tab feature

- **Slides Pane:** The Slides group is used to manage slides in a presentation. It allows users to create a new slide, choose different slide layouts, duplicate slides, delete slides, and reset slide formatting. This feature helps organize the structure and flow of the presentation.
- **Slides:** The Slides area displays thumbnails of all slides in the presentation. It allows users to easily navigate between slides, rearrange their order by dragging, and get an overview of the entire presentation. This helps in organizing and reviewing slides efficiently.
- **Font:** The Font group is used to format text on slides. It includes options to change the font type, size, color, and style such as bold, italic, and underline. Users can also apply text effects to make the content more clear and visually attractive.
- **Paragraph:** The Paragraph group controls the alignment and spacing of text. It allows users to align text left, right, center, or justify, adjust line spacing, create bullet points and numbering, and control text indentation. This helps improve the readability and presentation of text.

- **Drawing:** The Drawing group provides tools for inserting and formatting shapes. Users can draw shapes, apply fill colors, outlines, and effects, and arrange objects on slides. This feature is useful for creating diagrams, flowcharts, and visual elements.
- **Editing:** The Editing group includes tools such as Find, Replace, and Select. These tools help users quickly locate text, replace words or phrases, and select objects on a slide. This feature saves time and improves editing efficiency.

4.Insert

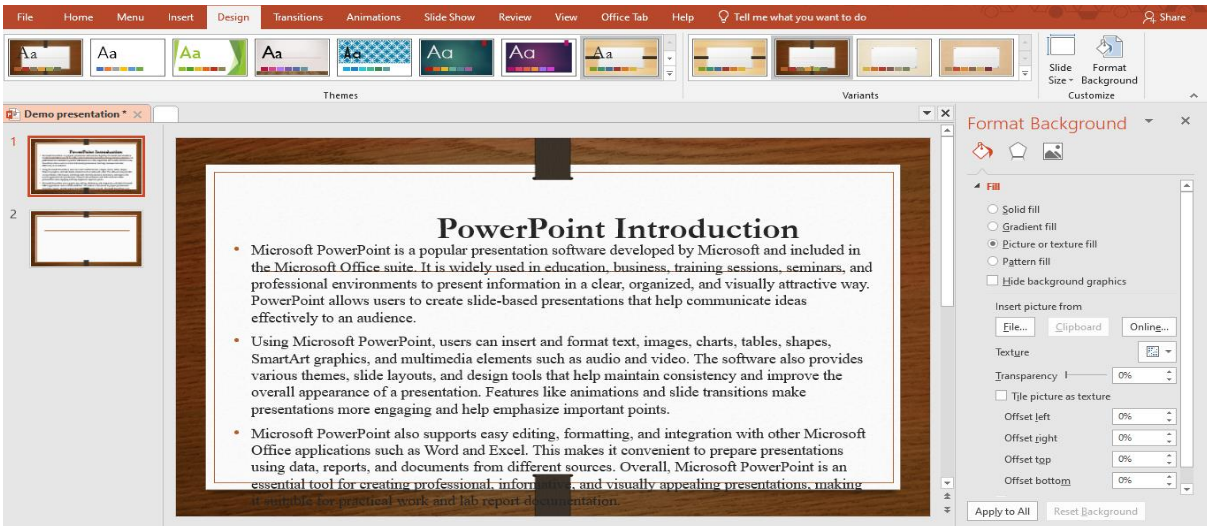
Insert Tab in Microsoft PowerPoint is used to add different types of content to a presentation. It allows users to insert text boxes, pictures, shapes, tables, charts, SmartArt graphics, icons, audio, and video into slides. The Insert tab helps enhance presentations by adding visual, graphical, and multimedia elements, making the content more informative, interactive, and visually appealing.



- **New Slide:** The New Slide feature is used to add a new slide to the presentation. Users can choose different slide layouts to organize content properly. This helps in structuring the presentation in a clear and logical way.
- **Tables:** The Tables feature allows users to insert tables into slides to organize data in rows and columns. Tables are useful for presenting information clearly, such as comparisons, schedules, or numerical data.
- **Images:** The Images feature is used to insert pictures into slides from a computer or other sources. Images help make presentations more attractive and support better visual understanding of the content.
- **Illustrations:** The Illustrations feature allows users to add shapes, icons, SmartArt, and charts. These visual elements are helpful for creating diagrams, flowcharts, and graphical representations of data.
- **Font:** The Font feature is used to format text on slides. It includes options to change font style, size, color, and text effects, helping improve readability and visual appeal.
- **Media:** The Media feature allows users to insert audio and video files into slides. This helps create interactive presentations by adding sound effects, narration, or video demonstrations.

5.Design

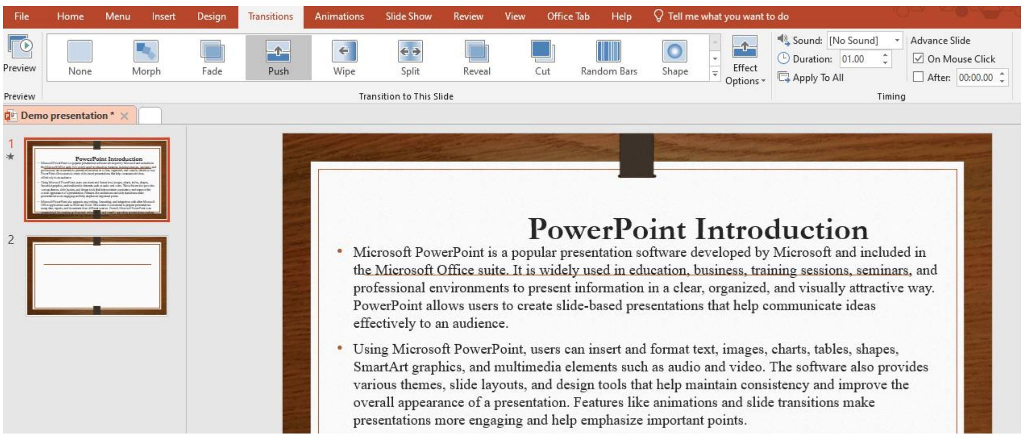
Design Tab in Microsoft PowerPoint is used to enhance the overall look and feel of a presentation. It provides tools to apply themes, choose slide backgrounds, adjust colors, fonts, and effects, and customize the slide size. The Design tab helps make presentations visually appealing, professional, and consistent in style.



- **Themes:** The Themes tool allows users to apply a set of coordinated colors, fonts, and effects to all slides, giving the presentation a consistent and professional look.
- **Variants:** The Variants tool lets users customize a selected theme by changing its color, font style, and effects to better suit the presentation's style.
- **Slide Size:** The Slide Size tool allows users to choose the dimensions of slides, such as standard (4:3) or widescreen (16:9), to fit different screens or printing formats.
- **Format Background:** The Format Background tool enables users to customize the background of slides with colors, gradients, textures, or images, enhancing visual appeal.

6.Transitions

The **Transition tab** in Microsoft PowerPoint is a powerful tool used to add visual effects when moving from one slide to another during a presentation. Transitions help make a presentation more dynamic and engaging by providing smooth or eye-catching slide changes, which can hold the audience's attention and improve the overall flow of the presentation. Using transitions effectively can make the content appear more professional and polished.

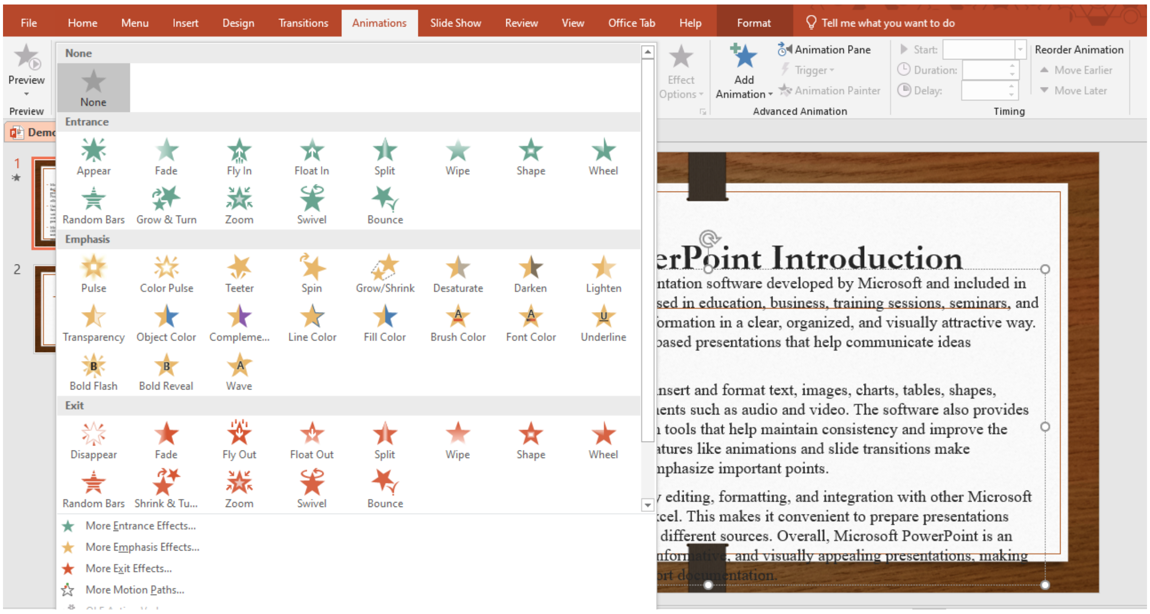


Key Features of the Transition Tab:

- **Transition Effects:** PowerPoint offers a wide range of transition effects such as Fade, Wipe, Push, Split, and Morph. Each effect provides a unique way for slides to appear or disappear, allowing presenters to choose the style that best suits their content and audience.
- **Effect Options:** Many transitions can be further customized using Effect Options. For example, you can change the direction of a Wipe or the style of a Split, giving more control over how the slides appear.
- **Duration:** This feature allows users to control how long each transition takes. By adjusting the duration, presenters can make transitions faster for a quick flow or slower to emphasize a particular slide.
- **Apply to All:** With this option, the selected transition effect can be applied to all slides in the presentation, ensuring visual consistency and saving time during the design process.
- **Advance Slide:** Users can set slides to advance manually, such as on a mouse click, or automatically after a specific time. This is especially useful for self-running presentations or timed demonstrations.

6.Animations

The **Animations tab** in Microsoft PowerPoint is used to apply motion effects to objects on a slide, such as text, images, shapes, charts, or SmartArt. Animations help make presentations more engaging by controlling how and when elements appear, move, or disappear on a slide. Proper use of animations can highlight important information, guide the audience's attention, and make the presentation more interactive and professional.



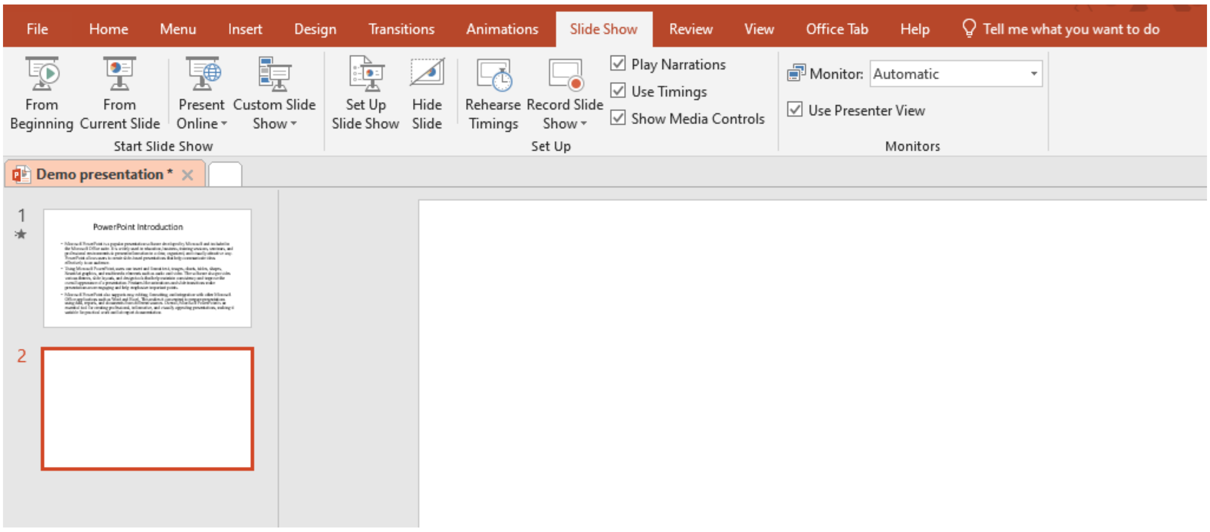
Key Features of the Animations Tab:

- **Animation Effects:** PowerPoint provides a variety of animation effects, including Entrance (objects appear on the slide), Emphasis (objects draw attention while on the slide), **Exit** (objects disappear from the slide), and **Motion Paths** (objects move along a defined path). These effects help present information dynamically.

- **Effect Options:** Many animations can be customized with Effect Options. Users can adjust the direction, sequence, or style of an animation, giving more control over how objects behave on the slide.
- **Animation Pane:** The Animation Pane allows users to view, manage, and reorder all animations on a slide. This helps organize complex animations and ensures smooth flow during the presentation.
- **Duration:** This feature controls how long an animation lasts. Adjusting the duration helps synchronize animations with the presenter's speech or other slide content.
- **Start Options:** Animations can be set to start On Click (manual), With Previous (simultaneous with another animation), or After Previous (automatically after another animation). These options help control timing and flow of content.
- **Reorder Animations:** Users can change the sequence of animations to ensure the proper order of information presentation.

7.Slide Show

The **Slide Show tab** in Microsoft PowerPoint is used to control how a presentation is delivered to the audience. It provides tools to start, customize, and rehearse slide shows, helping presenters deliver their content smoothly and professionally. This tab is essential for preparing and presenting slides effectively, ensuring that the audience experiences the presentation as intended.



Key Features of the Slide Show Tab:

- **From Beginning / From Current Slide:** These options allow users to start the slide show either from the first slide or from the currently selected slide. This is useful for practicing or presenting specific sections.
- **Custom Slide Show:** This feature allows users to create a customized sequence of slides for different audiences or purposes, without changing the main presentation.
- **Set Up Slide Show:** Users can configure how the slide show will run, including options for looping, showing slides in a specific order, or presenting on multiple monitors.

- **Rehearse Timings:** This tool helps presenters practice their presentation and record the time spent on each slide. It is useful for pacing and preparing for timed presentations.
- **Record Slide Show:** This feature allows users to record narration, slide timings, and annotations, creating a self-running presentation that can be shared with others.
- **Hide Slide:** Users can temporarily hide slides from the slide show without deleting them, allowing flexibility in presentations.
- **Play Narrations / Use Timings:** These options allow recorded narrations and slide timings to be included or excluded during the presentation, providing full control over delivery.

8.Conclusion

Microsoft PowerPoint is a versatile and powerful tool for creating professional and visually appealing presentations. By using its various tabs and features, such as Home, Insert, Design, Transitions, Animations, and Slide Show, users can effectively organize, format, and enhance content to communicate ideas clearly. PowerPoint allows the integration of text, images, charts, multimedia, and animation, making presentations more engaging and interactive. Understanding and utilizing these features helps users deliver information in a structured and visually appealing manner, improving both learning and communication outcomes. Overall, Microsoft PowerPoint is an essential application for academic, professional, and personal presentations, providing a platform to present ideas creatively and effectively.

Microsoft Excel

Excel is a commercial spreadsheet application produced and distributed by Microsoft for Microsoft Windows and Mac OS. It features the ability to perform basic calculations, use graphing tools, create pivot tables and create macros. Excel has the same basic features as all spreadsheet applications, which use a collection of cells arranged into rows and columns to organize and manipulate data. They can also display data as charts, histograms, and line graphs. Excel permits users to arrange data so as to view various factors from different perspectives. Visual Basic is used for applications in Excel, allowing users to create a variety of complex numerical methods.

Excel is typically used to organize data and perform financial analysis. It is used across all business functions and at companies from small to large.

The main uses of Excel include:

- Data entry
- Data management
- Accounting
- Financial analysis
- Charting and graphing
- Programming
- Time management
- Task management
- Financial modeling
- Customer relationship management (CRM)

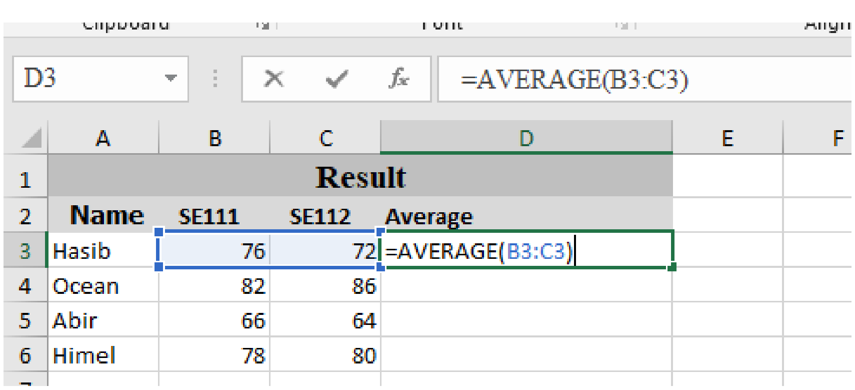
Excel Interface

Opening Excel reveals a spreadsheet with these key elements:

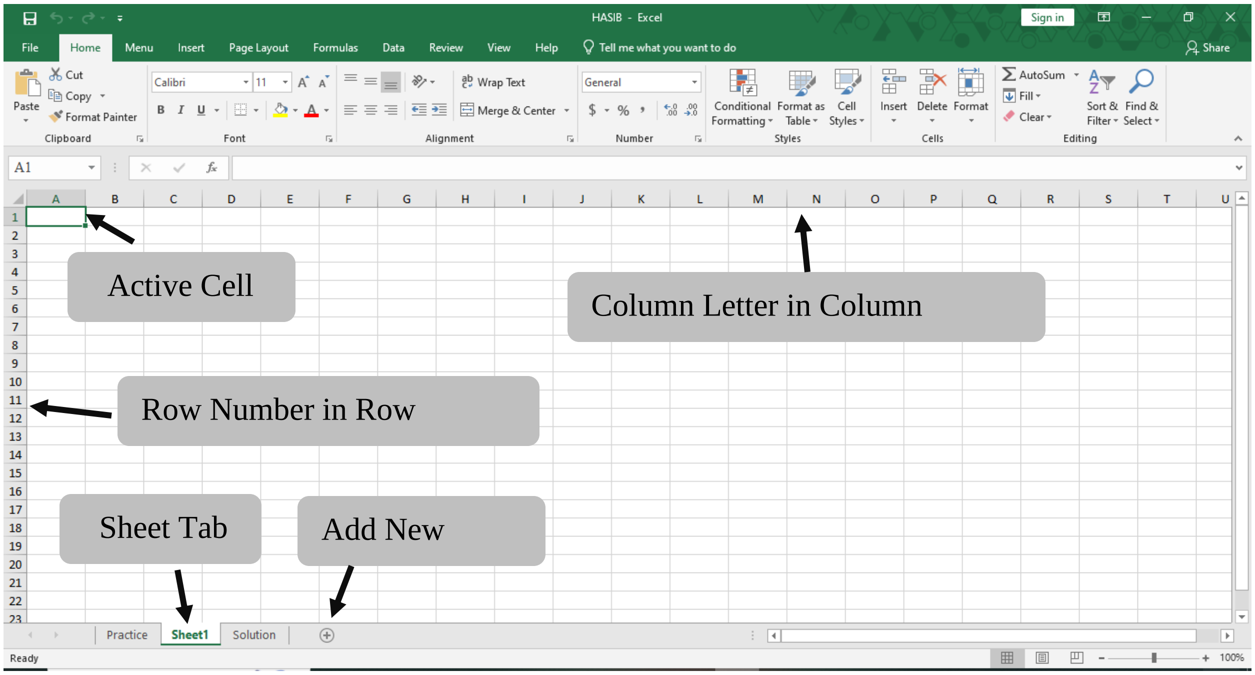
- **Ribbon:** The top toolbar with tabs like Home, Insert and Data. Each tab contains tools, such as formatting options or chart creation.

- **Quick Access Toolbar:** A small bar above the ribbon with shortcuts like Save or Undo.

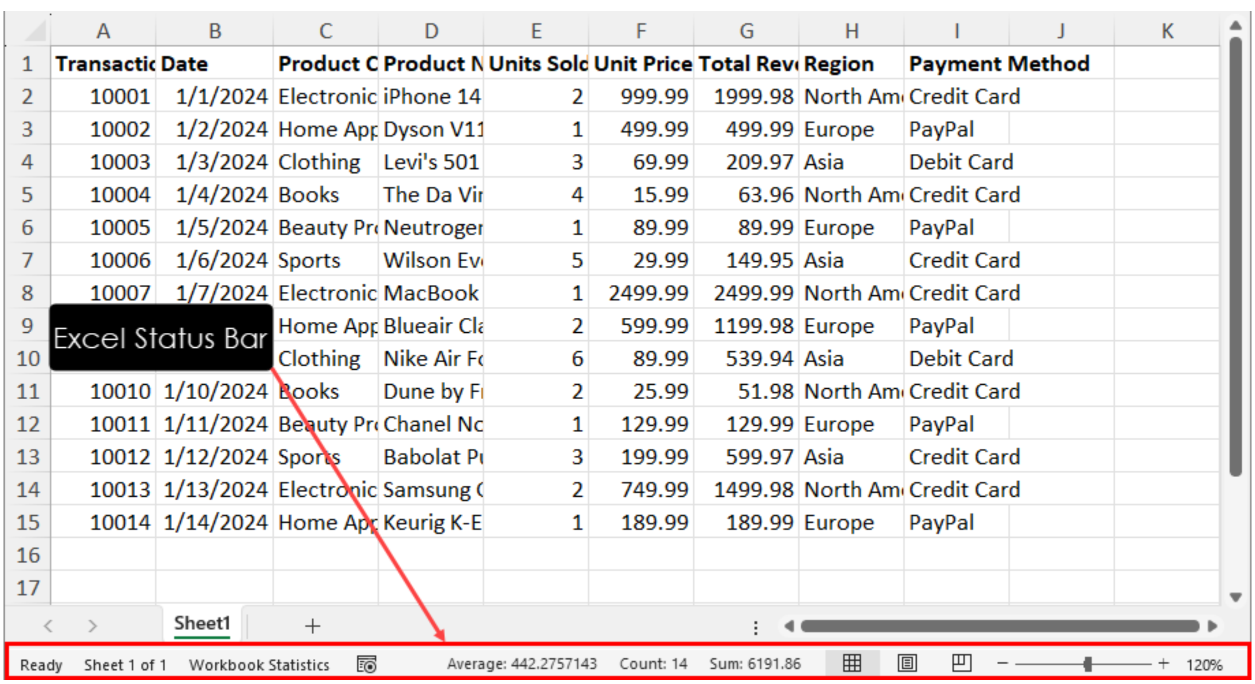
- **Formula Bar:** Above the worksheet, it displays the selected cell's content, such as text or formulas.



❖ **Worksheet:** The grid of rows (numbered) and columns (lettered). Cells, like A1 or B2, are where we input data.



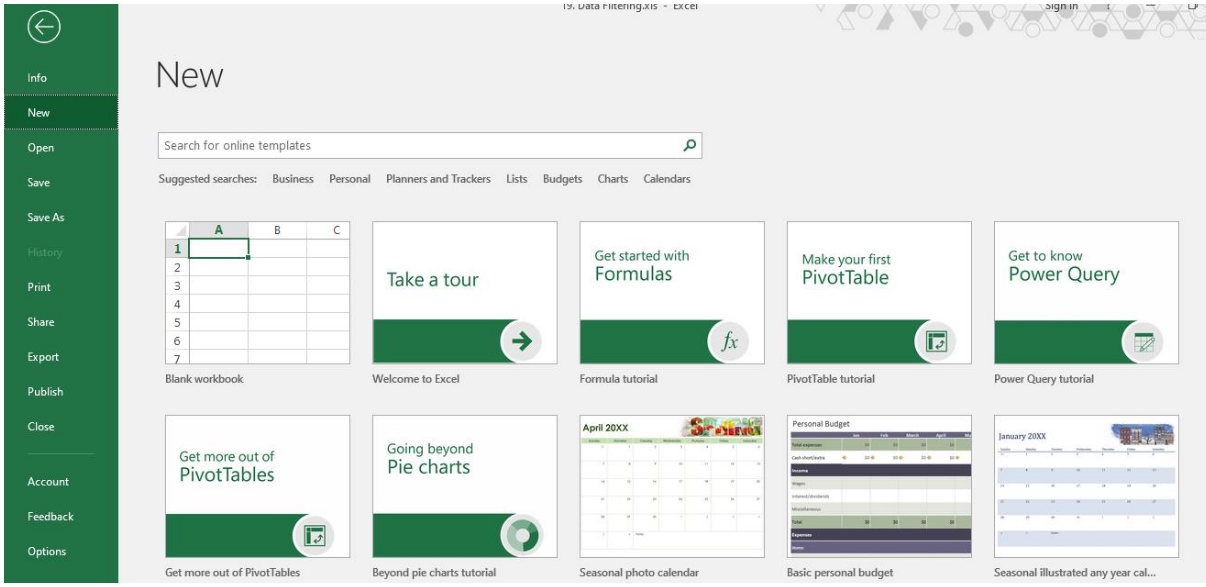
Status Bar: At the bottom, it shows quick calculations (e.g., sum or average) for selected cells.



	A	B	C	D	E	F	G	H	I	J	K
1	Transactic	Date	Product C	Product N	Units Solc	Unit Price	Total Rev	Region	Payment Method		
2	10001	1/1/2024	Electronic	iPhone 14	2	999.99	1999.98	North Am	Credit Card		
3	10002	1/2/2024	Home App	Dyson V11	1	499.99	499.99	Europe	PayPal		
4	10003	1/3/2024	Clothing	Levi's 501	3	69.99	209.97	Asia	Debit Card		
5	10004	1/4/2024	Books	The Da Vir	4	15.99	63.96	North Am	Credit Card		
6	10005	1/5/2024	Beauty Pr	Neutroger	1	89.99	89.99	Europe	PayPal		
7	10006	1/6/2024	Sports	Wilson Ev	5	29.99	149.95	Asia	Credit Card		
8	10007	1/7/2024	Electronic	MacBook	1	2499.99	2499.99	North Am	Credit Card		
9			Home App	Blueair Cl	2	599.99	1199.98	Europe	PayPal		
10			Clothing	Nike Air F	6	89.99	539.94	Asia	Debit Card		
11	10010	1/10/2024	Books	Dune by F	2	25.99	51.98	North Am	Credit Card		
12	10011	1/11/2024	Beauty Pr	Chanel Nc	1	129.99	129.99	Europe	PayPal		
13	10012	1/12/2024	Sports	Babolat P	3	199.99	599.97	Asia	Credit Card		
14	10013	1/13/2024	Electronic	Samsung	2	749.99	1499.98	North Am	Credit Card		
15	10014	1/14/2024	Home App	Keurig K-E	1	189.99	189.99	Europe	PayPal		
16											
17											

1.File

The File tab in Microsoft Excel provides access to commands used to manage workbooks. It allows users to create new files, open existing workbooks, save or save files with a new name, and print worksheets. The File tools also include options to share files, export data, protect workbooks, and adjust Excel settings. These tools help users control file-related tasks efficiently and manage their Excel documents properly.



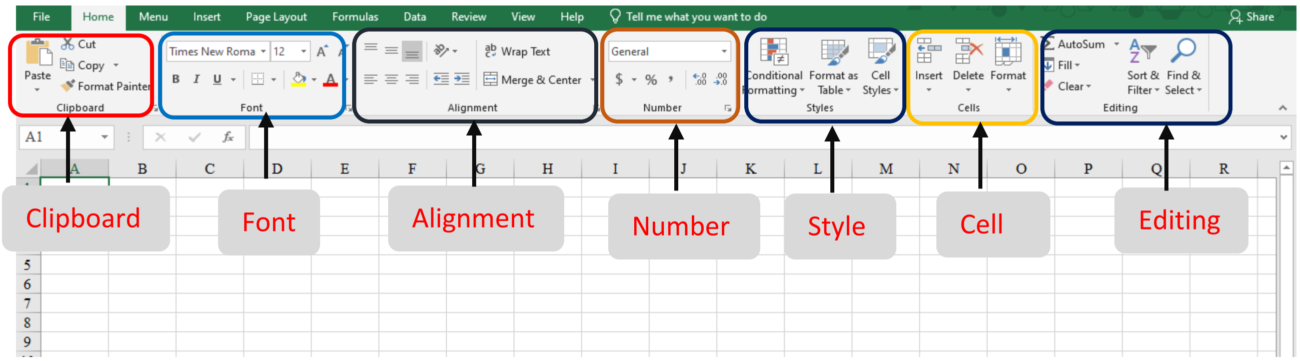
***File Tools*:**

Clicking the File tab opens the Backstage View for file management:

- **Info:** Protect Workbook, Check for Issues, Manage Workbook versions.
- **New:** Create a new blank workbook or use templates.
- **Open:** Access recent files or browse for others.
- **Save & Save As:** Save your work in various formats like .xlsx, .csv, .pdf.
- **Print:** Preview and print your spreadsheets.
- **Share:** Share via OneDrive, email, or present online.
- **Export:** Create PDF/XPS or change file types.
- **Account & Options:** Manage settings and account info.

2.Home

The Home tab in Microsoft Excel contains the most frequently used tools for creating, editing, and formatting worksheets. It provides options to change font type, size, color, and cell styles to improve the appearance of data. Users can align text, merge cells, wrap text, and adjust row height and column width for better layout. The Home tools also allow users to format numbers as currency, percentage, date, or decimal values. In addition, this tab includes tools for inserting, deleting, and clearing cells, sorting and filtering data, applying conditional formatting, and performing basic editing tasks such as copy, paste, and find and replace. These features help users organize data clearly and work more efficiently in Excel.



Key Components of the Excel Home Tab:

Clipboard: For Cut, Copy, Paste.

Font: Change font type, size, color, bold, italicize.

Alignment: Text alignment, merge cells, text orientation.

Number: Format cells as currency, percentage, dates, etc.

Styles: Conditional formatting, cell styles, format as table.

Cells: Insert/delete rows/columns, format cells.

Editing: AutoSum, Fill, Clear, Sort & Filter, Find & Select.

Excel Formulas

One of Excel's strengths lies in its ability to perform quick and accurate calculations using formulas and functions. Even in basic spreadsheets, we might use:

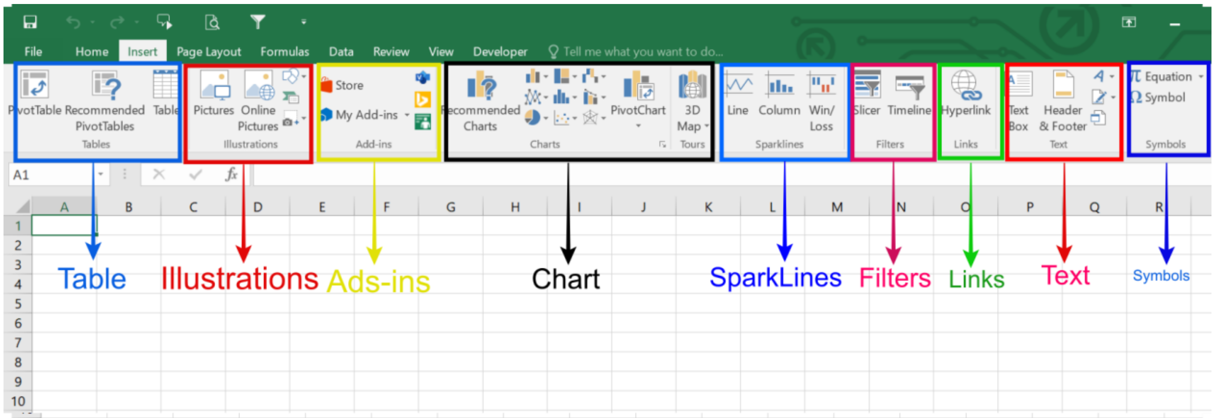
Formula	Purpose	Definition
SUM	Adds a range of numbers	Totals values in a specified range
AVERAGE	Calculates the mean of a range	Finds the average of values in a range
MAX	Finds the largest value in a range	Returns the highest value in a specified range
MIN	Finds the smallest value in a range	Returns the lowest value in a specified range

D9 =SUM(D3:D8)

	A	B	C	D
1	Budget Planning			
2	S/N	Item	Planned Cost	Actual Cost
3	1	Cloth	780	600
4	2	Shirt	670	500
5	3	Pant	750	650
6	4	Bag	1200	1000
7	5	Tie	500	400
8	6	Tie Pin	498	348
9		Total Cost	4398	3498
10		Agerage	733	583
11		Maximum	1200	1000
12		Minimum	498	348

3.Insert

The Insert tab in Microsoft Excel is used to add various objects and elements into a worksheet to enhance data presentation and analysis. It allows users to insert tables, charts, illustrations, links, text, and symbols that help visualize information more effectively. By using the tools available in the Insert tab, users can transform raw data into meaningful and interactive content. This tab plays an important role in making worksheets more organized, informative, and visually appealing. In the following section, key Insert tools and groups are discussed with the help of images.

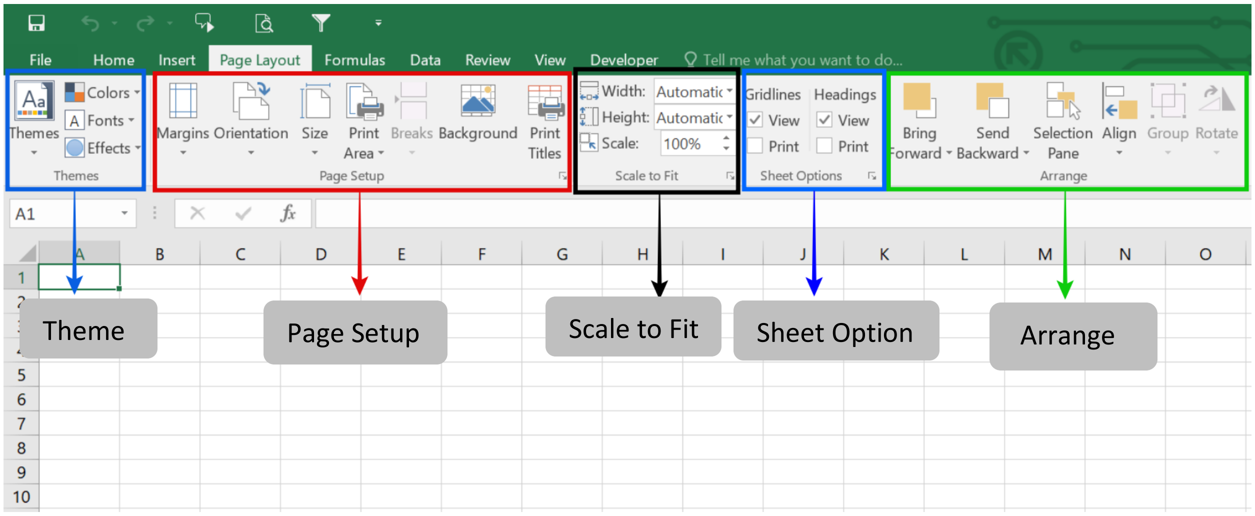


The image displays the "Insert" tab ribbon in Microsoft Excel, highlighting various tools for adding different elements to a spreadsheet.

- ***Table:*** Create and format tables to organize data.
- ***Illustrations:*** Insert pictures, shapes, icons, and 3D models.
- ***Add-ins:*** Access the Office Store to find and manage additional functionalities.
- ***Chart:*** Visualize data using a variety of chart types, including recommended options and PivotCharts.
- ***SparkLines:*** Add small charts within a single cell to show data trends.
- ***Filters:*** Utilize Slicers and Timelines for interactive data filtering.
- ***Links:*** Insert hyperlinks to connect to other locations or files.
- ***Text:*** Add text boxes, headers, and footers.
- ***Symbols:*** Insert mathematical equations or special symbols.

4.Page Layout

The Page Layout tab in Microsoft Excel is used to control the appearance and layout of a worksheet, especially for printing purposes. It allows users to adjust page settings such as margins, orientation, paper size, and scaling to ensure that data is printed clearly and professionally. This tab also provides tools for applying themes, changing page background, and organizing worksheet elements properly. By using the Page Layout tools, users can improve the overall structure and visual consistency of a worksheet. In the following section, the key Page Layout tools and groups are explained with the help of images.

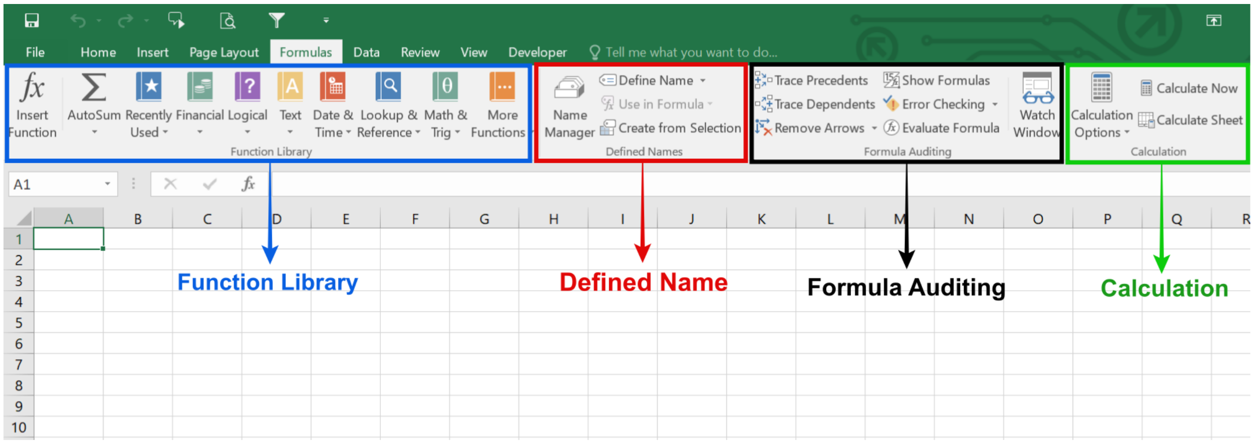


Key Groups in the Page Layout Tab:

- **Themes:** Quickly change the overall look with coordinated colors, fonts, and effects.
- **Page Setup:**
  - **Margins:** Adjust space around the content.
  - **Orientation:** Switch between Portrait (tall) and Landscape (wide).
  - **Size:** Select paper size (e.g., Letter, A4).
  - **Print Area:** Define which cells get printed.
  - **Breaks:** Insert or remove manual page breaks.
  - **Background:** Add a background image to the sheet.
  - **Print Titles:** Set rows or columns to repeat on every printed page.
- **Scale to Fit:** Resize content to fit a specific number of pages (e.g., "Fit to 1 page wide by 2 pages tall").
- **Sheet Options:** Control visibility of gridlines and headings (A, B, C; 1, 2, 3) on printouts.
- **Arrange:** Tools for positioning and formatting shapes, pictures, and other objects.

5.Formulas

The Formulas tab in Microsoft Excel is used to perform calculations and manage formulas efficiently within a worksheet. It provides access to a wide range of built-in functions such as mathematical, logical, financial, and statistical functions. This tab also includes tools for inserting functions, defining named ranges, checking formula errors, and auditing calculations. By using the Formulas tab, users can analyze data accurately and automate complex calculations. In the following section, the key Formula tools and groups are discussed with the help of images.

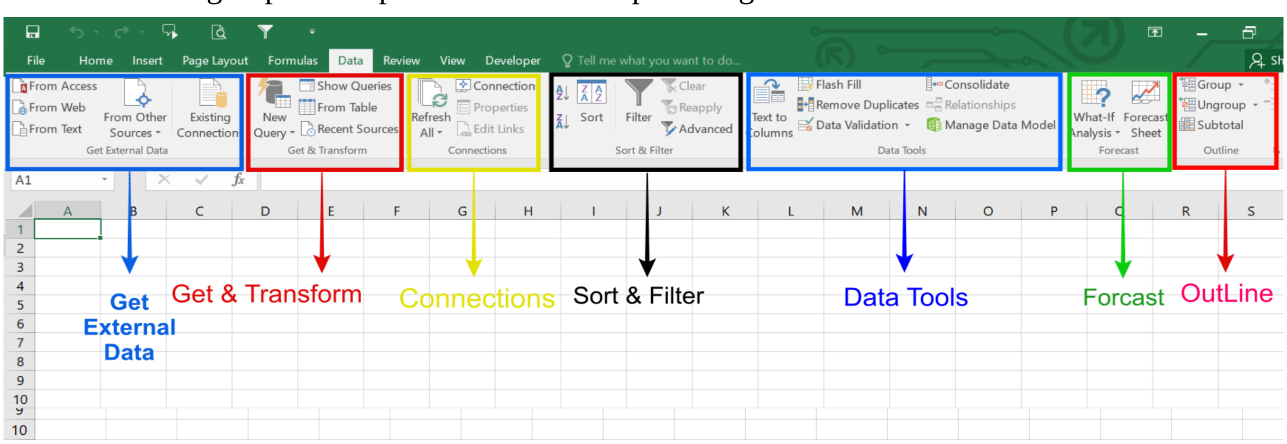


The image displays the **Formulas tab** in Microsoft Excel, highlighting the different groups of tools available for working with functions and calculations.

- **Function Library:** Contains a wide variety of built-in functions (e.g., Financial, Logical, Text, Date & Time, Lookup & Reference, Math & Trig) to help process data.
- **Defined Names:** Allows users to define, manage, and use names for cells or ranges within formulas.
- **Formula Auditing:** Provides tools like Trace Precedents, Trace Dependents, Error Checking, and Evaluate Formula to ensure formulas are correct.
- **Calculation:** Controls when and how calculations are performed in the workbook, with options like Calculate Now and Calculate Sheet.

6.Data

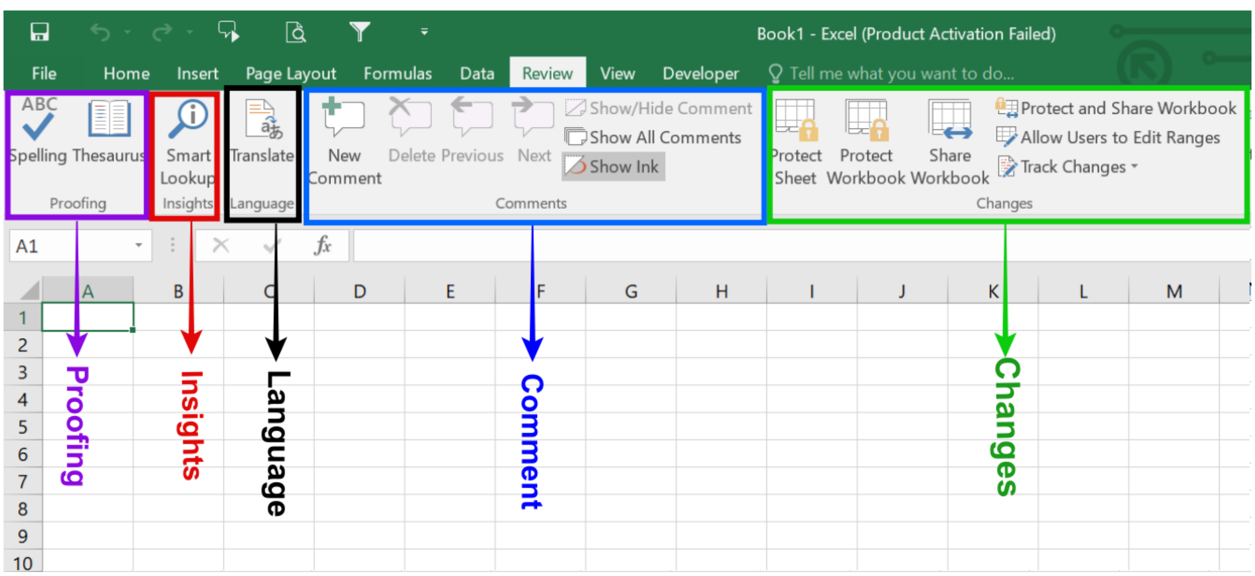
The Data tab in Microsoft Excel is used to manage, organize, and analyze large amounts of data effectively. It provides tools for importing data from different sources, sorting and filtering information, and cleaning data for accuracy. This tab also includes features for data validation, removing duplicates, and converting text into columns. In addition, the Data tab offers tools for analyzing data such as what-if analysis, forecasting, and grouping data. By using the Data tools, users can ensure data consistency and perform accurate analysis. In the following section, the key Data tools and groups are explained with the help of images.



- **Get External Data / Get & Transform:** This group is used to import data into Excel from external sources such as text files, databases, websites, or other Excel files. It also allows users to clean, transform, and organize imported data before using it in the worksheet.
- **Connections:** The Connections group manages links between Excel and external data sources. It allows users to refresh data, view existing connections, and control how external data is updated in the workbook.
- **Sort & Filter:** This group is used to arrange and analyze data easily. Users can sort data in ascending or descending order and apply filters to display only specific information based on selected criteria.
- **Data Tools:** The Data Tools group helps in cleaning and managing data. It includes features like Data Validation, Remove Duplicates, Text to Columns, and Flash Fill, which improve data accuracy and organization.
- **Forecast:** This group is used for predicting future trends based on existing data. It provides tools such as What-If Analysis and Forecast Sheet to help analyze possible outcomes.
- **Outline:** The Outline group helps organize large datasets by grouping and summarizing data. Users can collapse or expand rows and columns to view data in a structured and simplified way.

7.Review

The Review tab in Microsoft Excel provides tools for checking, reviewing, and protecting worksheets. It allows users to check spelling, add comments or notes, track changes, and protect workbooks and worksheets. These tools help improve accuracy, collaboration, and data security.



1. **Proofing:** The Proofing group is used to check spelling and grammar in worksheet content. It also provides tools like Thesaurus to find synonyms and improve text accuracy.

2. Insights: The Insights tool helps users analyze data automatically. It provides quick suggestions, patterns, and summaries from the selected data to support better decision-making.

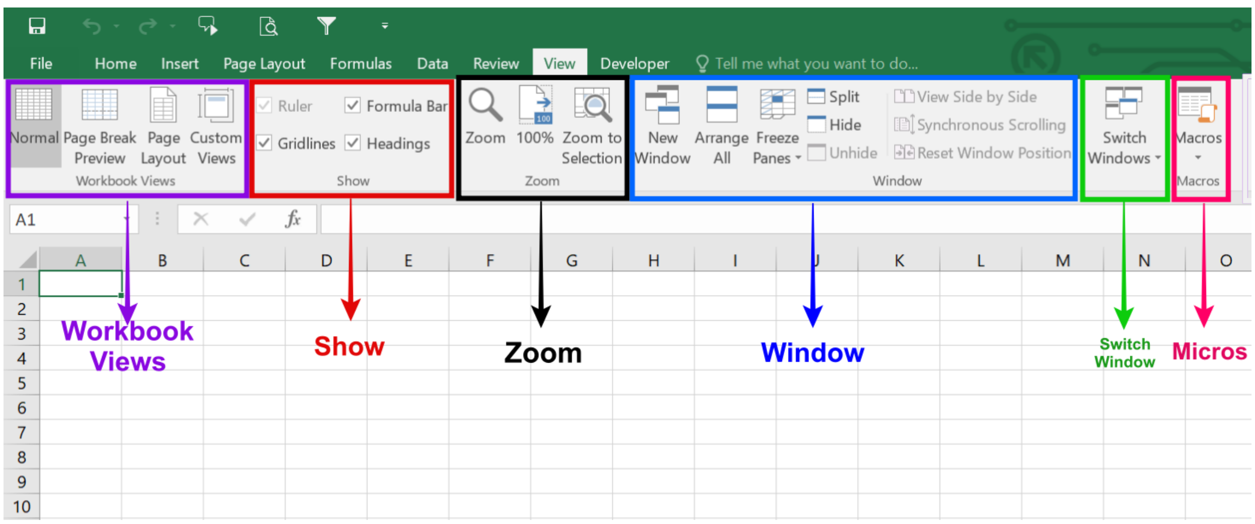
3. Language: The Language group is used to translate selected text into different languages. It also allows users to set the proofing language for spell checking.

4. Comments : The Comments group allows users to add, edit, delete, and navigate comments in a worksheet. It is mainly used for collaboration, feedback, and explanations without changing the actual data.

5. Changes: The Changes group is used to protect worksheets or workbooks and manage editing permissions. It also includes tools like Track Changes and Share Workbook to monitor and control modifications made by different users.

8.View

The View tab in Microsoft Excel is used to control how a worksheet is displayed on the screen. It allows users to switch between different workbook views, zoom in or out, show or hide gridlines and headings, freeze panes, and manage multiple windows. These tools help users view and work with data more comfortably and efficiently.



The marked items in the Microsoft Excel **View tab** image have the following functions:

Workbook Views: This group contains commands to change the overall display of the workbook, such as Normal, Page Break Preview, and Page Layout views.

Show: This group allows users to show or hide specific interface elements like gridlines, the formula bar, and column/row headings.

Zoom: This feature provides options to quickly zoom in, zoom out, or zoom to a specific selection on the worksheet.

Window: This group offers tools for managing open workbook windows, including opening a new window, arranging all open windows, splitting the view, or switching between different windows.

Macros: This functionality is used to record, view, or manage automated tasks (macros) within Excel.

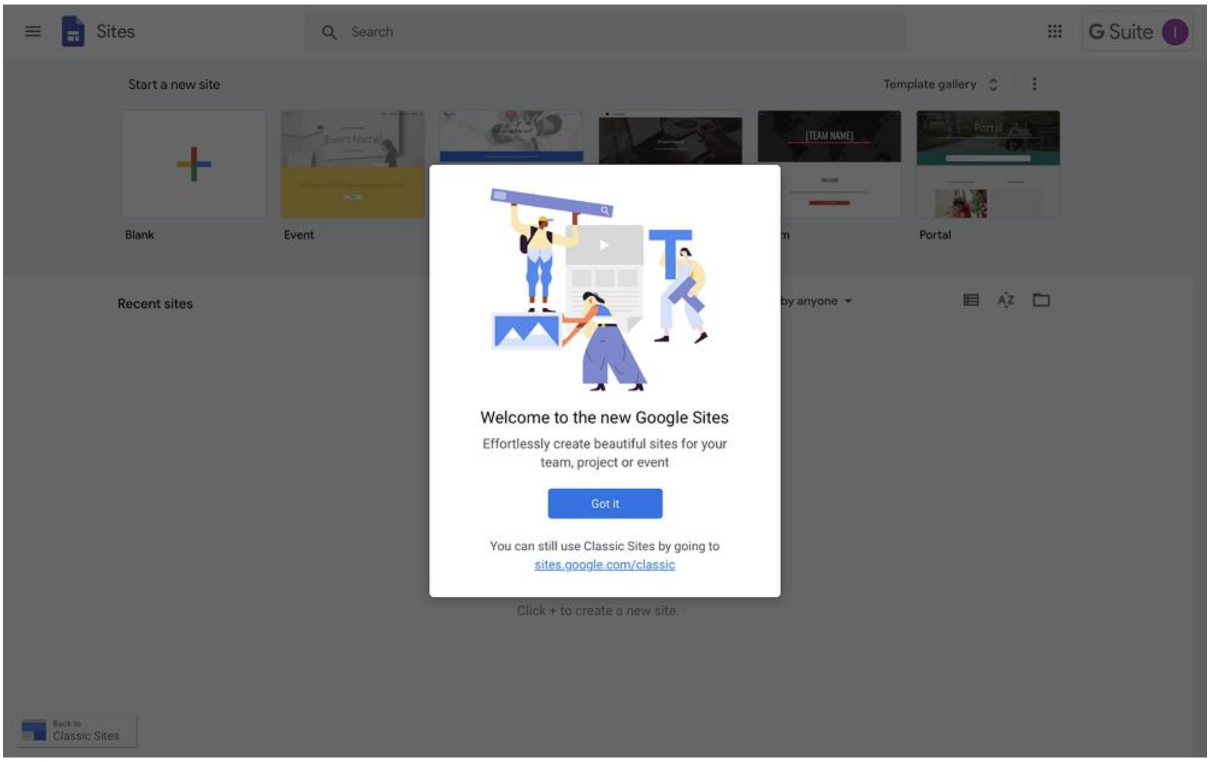
9.Conclusion

Microsoft Excel is a powerful and versatile application that plays a vital role in data management and analysis. With its wide range of features such as formulas, functions, charts, tables, and data tools, Excel enables users to organize information, perform accurate calculations, and present data in a clear and meaningful way. Whether used for academic work, business tasks, or personal projects, Microsoft Excel improves efficiency, accuracy, and productivity, making it an essential tool in today's digital world.

Google Sites

1. Introduction

Google Sites is a free website-building tool provided by Google. It allows users to create simple and professional websites without requiring any programming or web design skills. Google Sites is widely used for creating educational websites, lab reports, project pages, portfolios, and internal organizational sites. It is integrated with other Google services such as Google Drive, Google Docs, Google Slides, and Google Forms, which makes content management easy and efficient. Users can create, edit, and publish websites directly from a web browser.



Google Sites Features

- Google Sites comes with a host of useful features, including:
- Google Sites templates with responsive design across devices
- Themes that give a coherent, designed look
- The ability to embed Youtube videos, images, and other content
- Tables of contents and expandable text for easy overview
- Integration with other Google Workplace tools:
- Add documents or sheets (Google Docs and Google Sheets)
- Display forms (Google Forms)
- Add map (Google Maps)
- Show calendar (Google Calendar)
- Tool for embedding code to the pages
- User access and visibility management

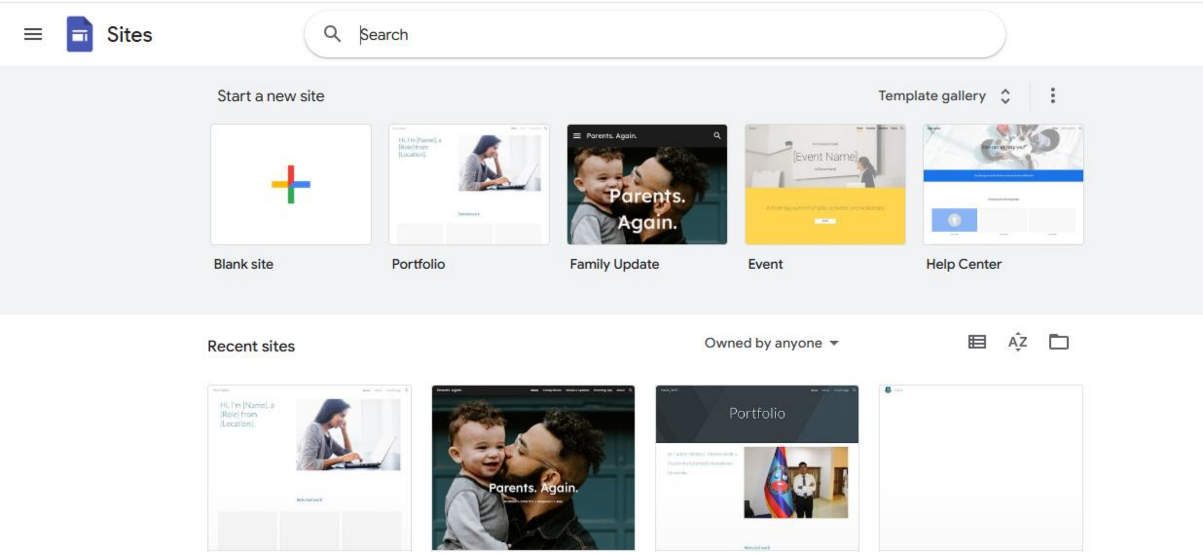
The Benefits of Google Sites

- Free to use
- Intuitive tools that let you get started in minutes, without installing or downloading any software
- You can access Google Sites anywhere and from any device
- Integrated with other Google Workplace tools such as Google Docs, Google Calendar, and Google Maps
- Easy to edit and manage access

2.Tools of Google Sites

A. Creating a New Google Site

Google Sites allows users to create a new website quickly. Users can start with a blank site or choose from pre-designed templates. The creation process is simple and requires only a Google account.



Steps:

- Open Google Sites
- Click on the Blank or any template
- A new editable site opens automatically

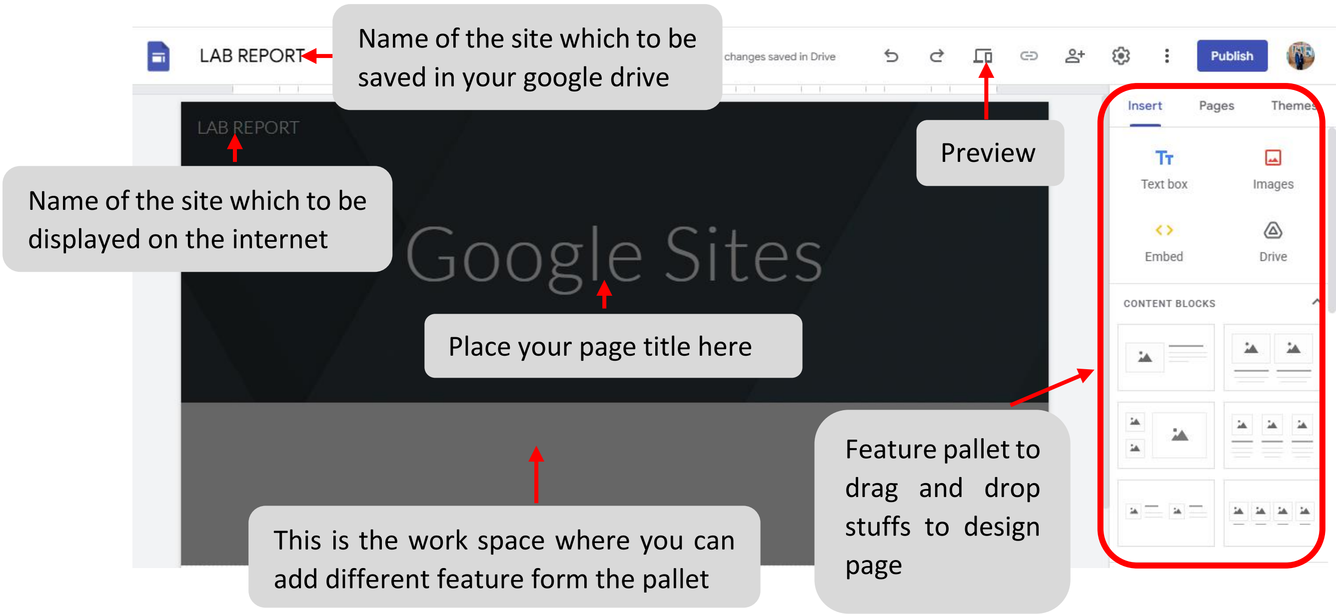
Use: Helps users start building a website easily.

B. Google Sites Interface

The Google Sites interface is user-friendly and designed for easy website creation. It includes a Header Area to display the site title and banner, an Insert Panel to add content like text, images, and videos, a Pages Panel to manage and organize different pages of the site, and a Themes Panel to customize the site's design and style. The Top Toolbar provides options to preview, share, and publish the site. Overall, the interface allows users to build and manage websites efficiently without requiring coding skills.

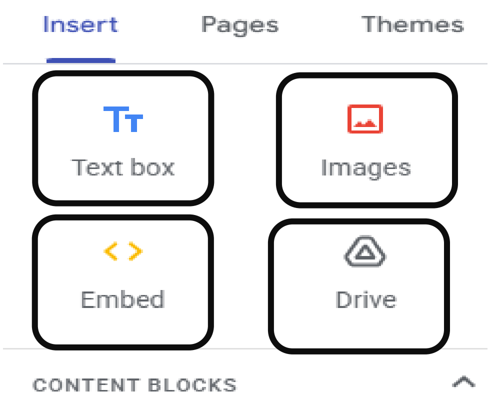
Main Parts of the Interface:

- Header Area – Displays site title and banner
- Insert Panel – Used to add content
- Pages Panel – Used to manage pages
- Themes Panel – Used to change design and style
- Top Toolbar – Contains preview, share, and publish options



C. Insert Tools

The Insert Tools in Google Sites allow users to add various types of content to their web pages. You can insert Text Boxes for writing content, Images from your computer or the web, Embed URLs or code, YouTube videos, Google Drive files (Docs, Sheets, Slides, etc.), and Buttons or Dividers to organize content. These tools make it easy to customize pages and present information in a clear and interactive way.



Features of Insert Tools and their uses

a) Text Box

The Text Box tool allows users to add written content such as descriptions, headings, and paragraphs.

Use:
Used to explain topics, write instructions, and add information.

b) Images

The Images tool allows users to upload images from a computer, Google Drive, or the web.

Use:
Used to add screenshots, diagrams, and photos to make the site more attractive and understandable.

c) Drive

The Drive option allows users to insert files directly from Google Drive such as Docs, Slides, Sheets, and Forms.

Use:
Helpful for embedding reports, presentations, and forms.

d) Embed

The Embed tool allows users to add content using a URL or embed code.

Use:
Used to embed videos, maps, charts, or external websites.

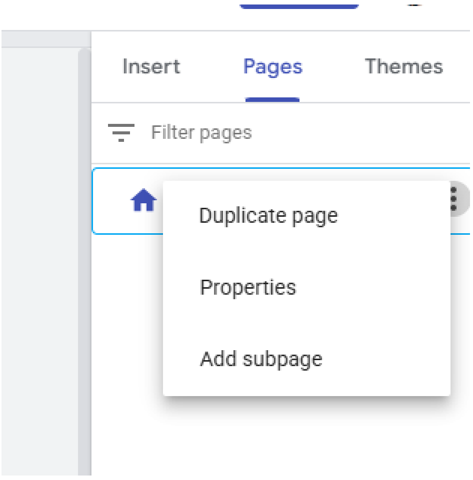
e) Layouts

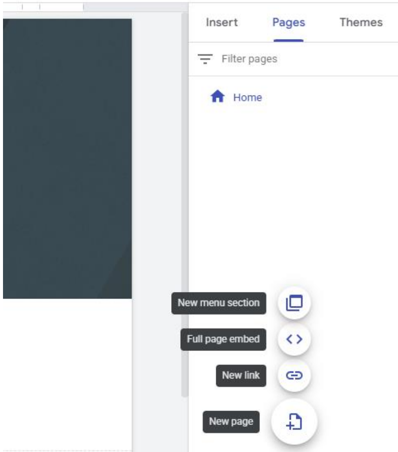
Layouts provide pre-designed structures to arrange text and images neatly.

Use:
Helps maintain a clean and organized website design.

D. Pages Features

The Pages Feature in Google Sites allows users to create, manage, and organize different pages of a website. You can add new pages, reorder pages, nest pages under other pages to create subpages, and rename or delete pages as needed. This feature helps structure the website, making navigation simple and intuitive for visitors.





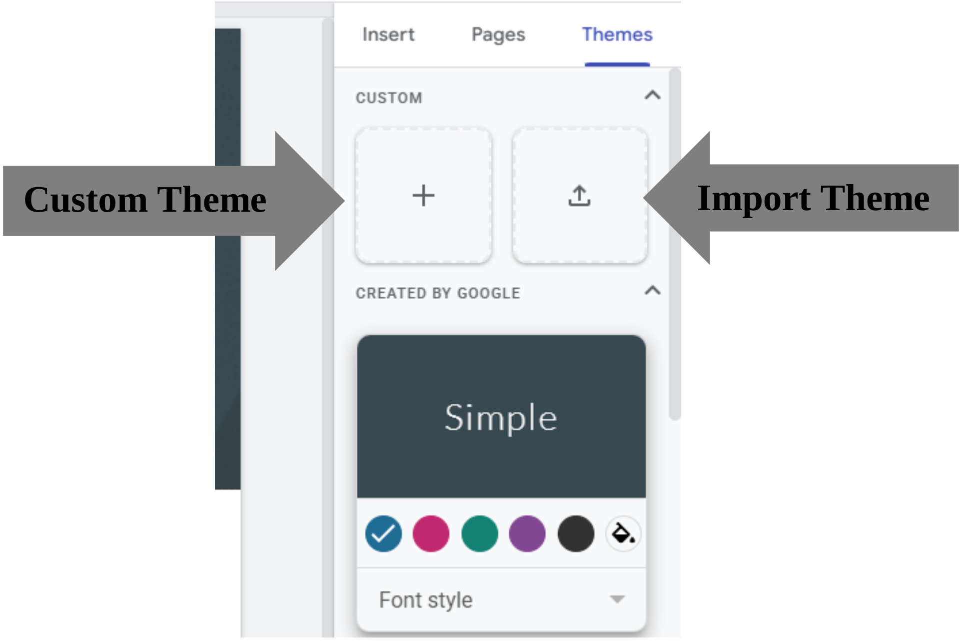
Functions:

- Add new pages
- Create subpages
- Rename or delete pages

Use:
Helps organize content in a structured manner.

E. Theme

The Themes Feature in Google Sites allows users to customize the overall design and appearance of their website. You can choose from a variety of pre-designed themes, adjust colors, fonts, and styles, and apply them consistently across all pages. This feature ensures a professional and visually appealing look, making the website more attractive and easier to navigate for visitors.



Features:

- Change font style
- Change color scheme
- Apply consistent design

Use:
Makes the website visually attractive and professional.

F. Preview Option

The Preview option in Google Sites allows you to see how your website will appear to visitors before publishing it. By clicking the Preview button, you can view your site in different device modes, such as desktop, tablet, and mobile, ensuring that the layout, text, images, and other elements are displayed correctly on all screens. This feature helps you check the design, alignment, and overall user experience, so you can make adjustments if needed before making your site public.

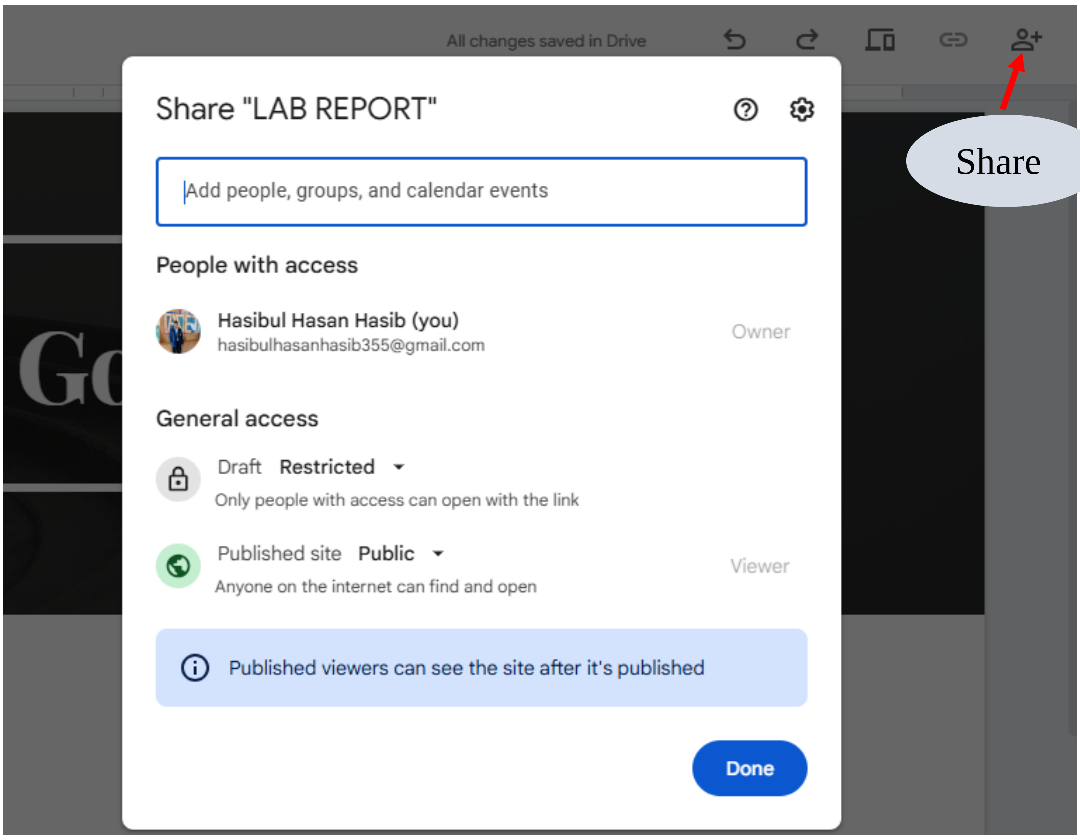
Preview Modes:

- Desktop
- Tablet
- Mobile

Use:
Helps ensure the website looks good on all screen sizes.

G. Share and Permissions

The Share and Permissions feature in Google Sites allows you to control who can view or edit your website. By clicking this option, you can invite specific people via their email addresses and assign them different roles, such as Viewer (can only see the site) or Editor (can make changes to the site). You can also set your site's visibility to the public, anyone with the link, or restrict it to specific people. This feature ensures your website is shared safely and only with the intended audience.



Permission Types:

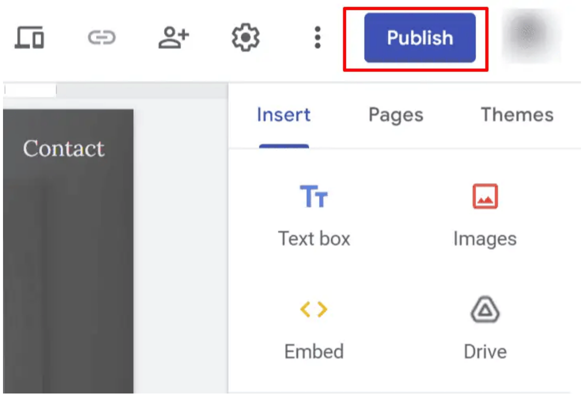
- Viewer
- Editor

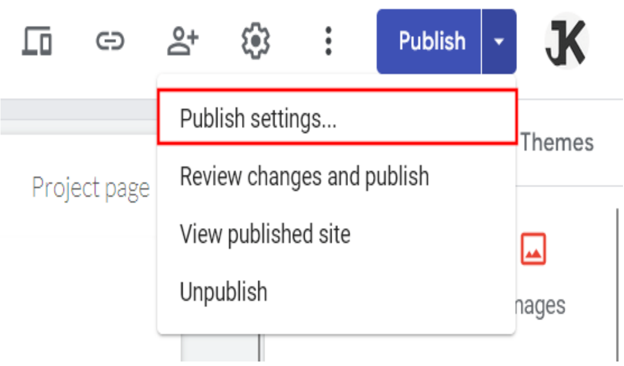
Use:

Ensures secure collaboration and controlled access.

H. Publish

The Publish option in Google Sites allows you to make your website live on the internet so that others can access it. By clicking the Publish button, you can choose a web address (URL) for your site and decide who can view it—either the public, anyone with the link, or specific people. Once published, your site is accessible to visitors, and any updates you make can be republished to reflect the latest changes. This feature is essential for sharing your work and making your website officially available online.





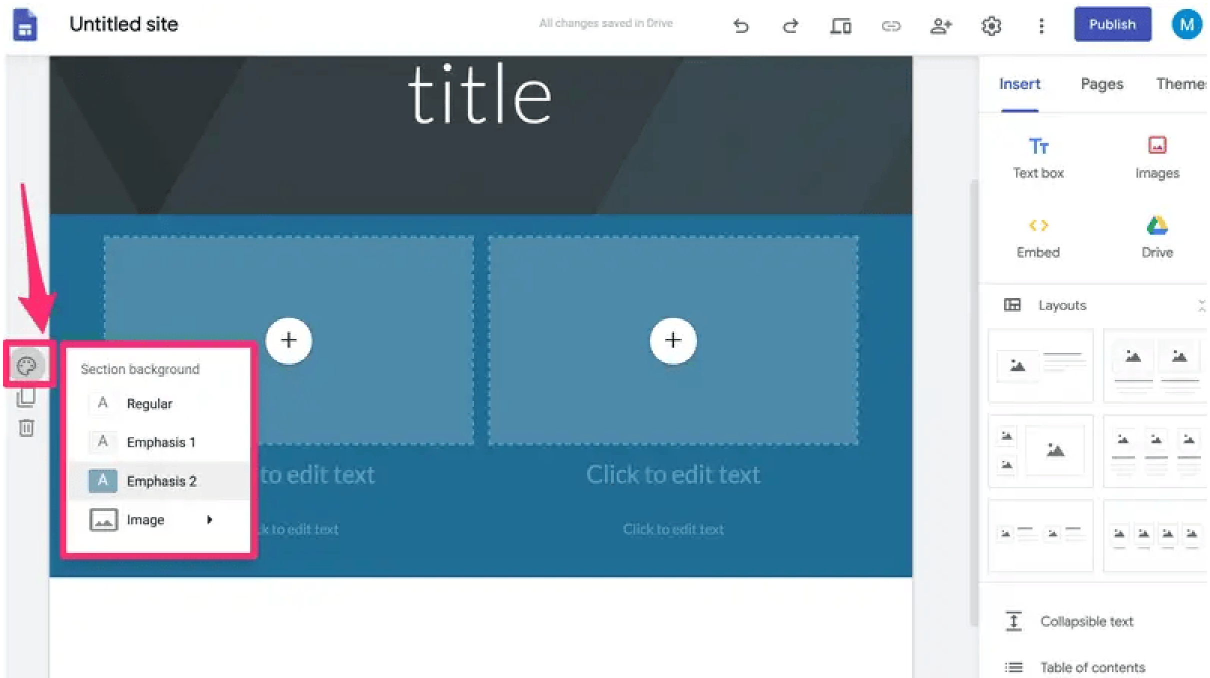
Steps:

- Click the Publish button
- Enter web address
- Set visibility options
- Click Publish

Use:
Allows others to access the website online.

I. Editing and Updating Site

The Editing and Updating feature in Google Sites allows you to make changes to your website even after it has been published. You can add or remove content, modify text, insert images, videos, or other elements, and adjust the layout as needed. After making updates, you can republish the site to ensure that visitors see the latest version. This feature ensures that your website stays current, accurate, and visually appealing over time.



J. Conclusion

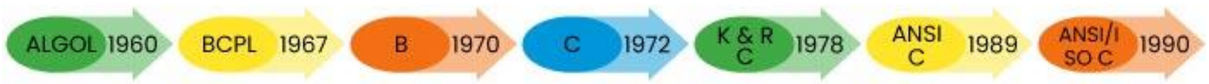
Google Sites is a powerful yet easy-to-use website creation tool. It helps users create professional websites without any coding knowledge. With features like Insert tools, Pages, Themes, Preview, and Publish options, users can design well-organized and attractive websites. Integration with Google Drive makes content management efficient. Therefore, Google Sites is an excellent platform for students, teachers, and professionals to create educational and project-based websites.

C Programming

C is a general-purpose procedural programming language initially developed by Dennis Ritchie in 1972 at Bell Laboratories of AT&T Labs. It was mainly created as a system programming language to write the UNIX operating system.

The history of the C programming language is quite fascinating and pivotal in the development of computer science and software engineering.

Year wise development of programming is as follows –



Why C Matters

- It is one of the most popular programming languages in the world.
- If you know C, you will have no problem learning other popular programming languages such as Java, Python, C++, C#, etc., as the syntax is similar.
- If you know C, you will understand how computer memory works.
- C is very fast, compared to other programming languages, like Java and Python.
- C is very versatile; it can be used in both applications and technologies.

Main features of C Language

- General Purpose and portable
- Procedural Programming Language
- Low-level Memory Access
- Fast Speed
- Clean Syntax

Application of C Language

- Gaming Engines
- Real Time Systems
- Embedded Systems
- Networking
- DBMS
- Medical devices
- Operating System

IDE for writing C code

An IDE (*Integrated Development Environment*) is used to edit and compile the code. Popular IDE's include **Code::Blocks**, **Eclipse**, and **Visual Studio**. These are all free, and they can be used to both edit and debug C code.

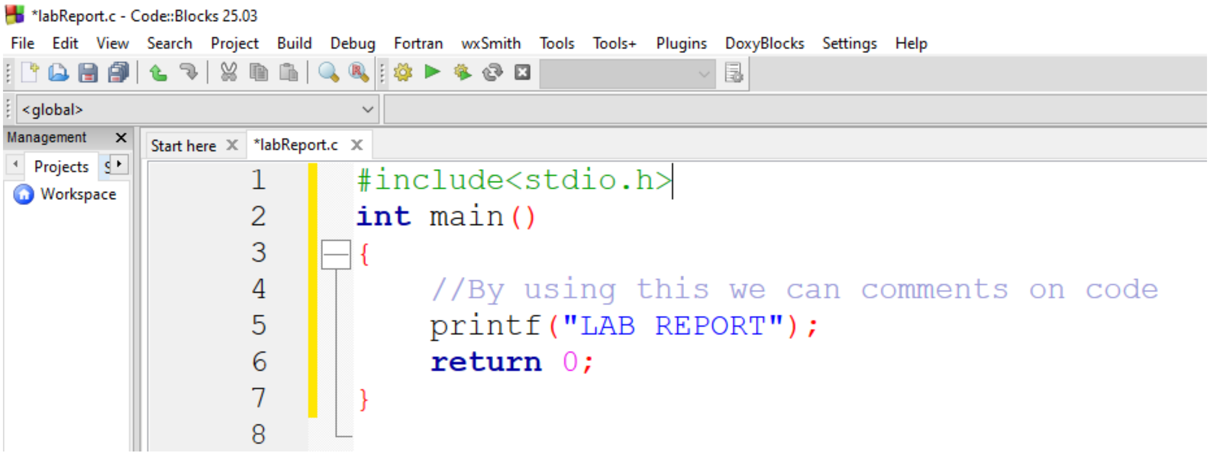
Here we will use CODEBLOCKS for writing our desired C program in our computer.

Open Codeblocks and go to *File* > *New* > *Empty File*.

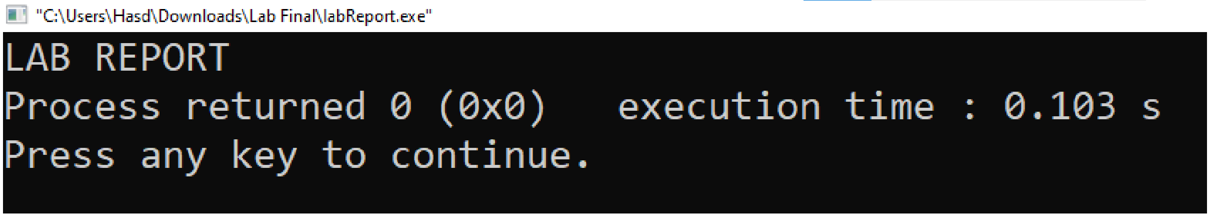
Write the following C code and save the file as ***labReport.c*** (*File* > *Save File as*):

Writing First Program in C Language

This simple program demonstrates the basic structure of a C program. It will also help us understand the basic syntax of a C program.



Then, go to **Build > Build and Run** to run (*execute*) the program. The result will look something to this:



Structure of the C program

After the above discussion, we can formally assess the basic structure of a C program. By structure, it is meant that any program can be written in this structure only. Writing a C program in any other structure will lead to a **Compilation Error**. The structure of a C program is as follows:

	1	#include<stdio.h>	Header
	2	int main()	Main
BODY	**3**	{	
	4	//by using this we can comment on code	Comment
	5	**printf**("LAB REPORT");	Statement
	6	return 0;	Return
	7	}	

Header Files Inclusion - Line 1 [#include <stdio.h>]

The first component is the Header files in a C program. A header file is a file with extension **.h** which contains C function declarations and macro definitions to be shared between several source files. All lines that start with **#** are processed by a preprocessor which is a program invoked by the compiler. In the above example, the preprocessor copies the preprocesses code of **stdio.h** to our file. The .h files are called header files in C.

Some of the C Header files:

- ***stddef.h*** - Defines several useful types and macros.
- ***stdint.h*** - Defines exact width integer types.
- ***stdio.h*** - Defines core input and output functions
- ***stdlib.h*** - Defines numeric conversion functions, pseudo-random number generator, and memory allocation
- ***string.h*** - Defines string handling functions
- ***math.h*** - Defines common mathematical functions.

Main Method Declaration - Line 2 [int main()]

The next part of a C program is the main() function. It is the entry point of a C program and the execution typically begins with the first line of the main(). The empty brackets indicate that the main doesn't take any parameter (. The int that was written before the main indicates the return type of main(). The value returned by the main indicates the status of program termination.

Body of Main Method - Line 3 to Line 6 [enclosed in {}]

The body of the main method in the C program refers to statements that are a part of the main function. It can be anything like manipulations, searching, sorting, printing, etc. A pair of curly brackets define the body of a function. All functions must start and end with curly brackets.

Comment - Line 4[// This prints "By using this we can comments on code"]

The comments are used for the documentation of the code or to add notes in your program that are ignored by the compiler and are not the part of executable program .

Statement - Line 4 [printf("LAB REPORT");]

Statements are the instructions given to the compiler. In C, a statement is always terminated by a semicolon (;). In this particular case, we use printf() function to instruct the compiler to display "LAB REPORT" text on the screen.

Return Statement - Line 5 [return 0;]

The last part of any C function is the return statement. The return statement refers to the return values from a function. This return statement and return value depend upon the return type of the function. The return statement in our program returns the value from main(). The returned value may be used by an operating system to know the termination status of your program. The value 0 typically means successful termination.

Identifiers in C

The names used to identify variables, functions, arrays, structures, and other user-defined objects in C programming are known as identifiers. A program element is uniquely identified by its name, which can be used to refer to it at a later stage of the program.

Example:

art here X *Hasib.c X

```
// Creating a variable
int val = 10;

// Creating a function
void func() {}

```

"val" and "func" are identifiers in the code sample above.

Rules for Naming Identifiers in C

A programmer must follow a set of rules to create an identifier in C:

- Identifier can contain following characters:
  - Uppercase (A-Z) and lowercase (a-z) alphabets.
  - Numeric digits (0-9).
  - Underscore (_).
- The first character of an identifier must be a letter or an underscore.
- Identifiers are case-sensitive.
- Identifiers cannot be keywords in C (such as int, return, if, while etc.).

Naming Conventions

In C programming, naming conventions are not strict rules but are commonly followed suggestions by the programming community for identifiers to improve readability and understanding of code. Below are some conventions that are commonly used:

For Variables:

- Use camelCase for variable names (e.g., frequencyCount, personName).
- Constants can use UPPER_SNAKE_CASE (e.g., MAX_SIZE, PI).
- Start variable names with a lowercase letter.
- Use descriptive and meaningful names.

For Functions:

- Use camelCase for function names (e.g., getName(), countFrequency()).
- Function names should generally be verbs or verb phrases that describe the action.

C VARIABLE

A variable in C is a named piece of memory which is used to store data and access it whenever required.

- It allows us to use the memory without having to memorize the exact memory address.
- To create a variable in C, we have to specify a name and the type of data it is going to store.
- C provides different data types that can store almost all kinds of data. For example, int, char, float, double, etc.
- Every variable must be declared before it is used. We can also declare multiple variables of same data type in a single statement by separating them using comma.

Data Type	Size	Description	Example
int	2 or 4 bytes	Stores whole numbers, without decimals	1
float	4 bytes	Stores fractional numbers, containing one or more decimals. Sufficient for storing 6-7 decimal digits	1.99
double	8 bytes	Stores fractional numbers, containing one or more decimals. Sufficient for storing 15 decimal digits	1.99
char	1 bytes	Stores a single character/letter/number, or ASCII values	'A'

Rules for Naming Variables in C

- We can assign any name to a C variable as long as it follows the following rules:
- A variable name must only contain letters, digits, and underscores.
- It must start with an alphabet or an underscore only. It cannot start with a digit.
- No white space is allowed within the variable name.
- A variable name must not be any reserved word or keyword.
- The name must be unique in the program.

Start here *Hasib.c

```c
#include <stdio.h>

int main()
{
    int age = 20;// integer variable
    float height = 5.7;// floating-point variable
    char grade = 'A'; // character variable

    printf("Age: %d\n", age);//%d Format Specifiers for char
    printf("Height: %.1f\n", height);//%f Format Specifiers for float
    printf("Grade: %c\n", grade);//%c Format Specifiers for char

    return 0;
}
```

OUTPUT:

```
Age: 20
Height: 5.7
Grade: A

Process returned 0 (0x0)   execution time : 0.052 s
Press any key to continue.
```

Basic Input and Output in C

In C, there are many input and output for different situations, but the most commonly used functions for Input/Output are ***scanf()*** and ***printf()*** respectively. These functions are part of the standard input/output library <stdio.h>. scanf() takes user inputs (typed using keyboard) and printf() displays output on the console or screen.

Basic Output in C

The printf() function is used to print formatted output to the standard output stdout (which is generally the console screen). It is one of the most commonly used functions in C.

The following examples demonstrate the use of printf for output in different cases:

Printing Some Text:

Start here × Hasib.c ×

```c
#include <stdio.h>

int main() {

    // Prints some text
    printf("Mr. K. M. Shahriar Islam");

    return 0;
}

```

OUTPUT:

```
Mr. K. M. Shahriar Islam
Process returned 0 (0x0)   execution time : 0.056 s
Press any key to continue.
```

Explanation: The text inside "" is called a string in C. It is used to represent textual information. We can directly pass strings to the printf() function to print them in console screen.

Basic Input in C:

scanf() is used to read user input from the console. It takes the format string and the addresses of the variables where the input will be stored.

Syntax

scanf("formatted_string", address_of_variables/values);

This function takes the address of the arguments where the read value is to be stored.

Examples of Reading User Input

Start here Hasib.c

```c
#include <stdio.h>

int main() {
    int student_id;
    printf("Enter your student_id: ");

    // Reads an integer
    scanf("%d", &student_id);

    // Prints the student id
    printf("Student Id is: %d\n", student_id);
    return 0;
}

```

OUTPUT:

"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"

```
Enter your student_id: 253
Student Id is: 253
```

Explanation: %d is used to read an integer; and &student_id provides the address of the variable where the input will be stored.

Operators in C

Operators are the basic components of C programming. They are symbols that represent some kind of operation, such as mathematical, relational, bitwise, conditional, or logical computations, which are to be performed on values or variables. The values and variables used with operators are called operands.

Unary, Binary and Ternary Operators

On the basis of the number of operands they work on, operators can be classified into three types:

1. Unary Operators: Operators that work on single operand.
   Example: Increment(++) , Decrement(--)
2. Binary Operators: Operators that work on two operands.
   Example: Addition (+), Subtraction(-) , Multiplication (*)
3. Ternary Operators: Operators that work on three operands.
   Example: Conditional Operator(? :)

Types of Operators in C

C language provides a wide range of built in operators that can be classified into 6 types based on their functionality:

1. Arithmetic Operators
2. Relational Operators
3. Logical Operator
4. Bitwise Operators
5. Assignment Operators
6. Other Operators (sizeof Operator, conditional operator)

Arithmetic Operators

The arithmetic operators are used to perform arithmetic/mathematical operations on operands. There are 9 arithmetic operators in C language:

Start here *Hasib.c

```c
#include <stdio.h>

int main() {

    int a = 25, b = 5;

    // using operators and printing results
    printf("a + b = %d\n", a + b);
    printf("a - b = %d\n", a - b);
    printf("a * b = %d\n", a * b);
    printf("a / b = %d\n", a / b);
    printf("a %% b = %d\n", a % b);
    printf("+a = %d\n", +a);
    printf("-a = %d\n", -a);
    printf("a++ = %d\n", a++);
    printf("a-- = %d\n", a--);

    return 0;
}
```

OUTPUT:

```
"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"
a + b = 30
a - b = 20
a * b = 125
a / b = 5
a % b = 0
+a = 25
-a = -25
a++ = 25
a-- = 26
```

<table>
<tr><th>Symbol</th><th>Operator</th><th>DESCRIPTON</th><th>SYNTAX</th></tr>
<tr><td>&&</td><td>Logical AND</td><td>Returns true if both the operands are true.</td><td>a && b</td></tr>
<tr><td>||</td><td>Logical OR</td><td>Returns true if both or any of the operand is true.</td><td>a || b</td></tr>
<tr><td>!</td><td>Logical NOT</td><td>Returns true if the operand is false.</td><td>a | b</td></tr>
<tr><td>=</td><td>Simple Assignment</td><td>Assign the value of the right operand to the left operand</td><td>a = b</td></tr>
<tr><td>+=</td><td>Plus and assign</td><td>Add the right operand and left operand and assign this value to the left operand.</td><td>a += b</td></tr>
</table>

Decision Making in C

In C, programs can choose which part of the code to execute based on some condition. This ability is called decision making and the statements used for it are called conditional statements. These statements evaluate one or more conditions and make the decision whether to execute a block of code or not.

1. if in C

The if statement is the simplest decision-making statement. It is used to decide whether a certain statement or block of statements will be executed or not i.e if a certain condition is true then a block of statements is executed otherwise not.

A condition is any expression that evaluates to either a true or false (or values convertible to true or flase).

```c
#include <stdio.h>

int main() {

    int age = 20;

    // If statement
    if (age >= 18) {
        printf("Eligible for vote");
    }
}
```

OUTPUT:

```
"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"
Eligible for vote
Process returned 0 (0x0)   execution time : 0.050 s
Press any key to continue.
```

The expression inside () parenthesis is the condition and set of statements inside {} braces is its body. If the condition is true, only then the body will be executed.

2. if-else in C

The if statement alone tells us that if a condition is true, it will execute a block of statements and if the condition is false, it won’t. But what if we want to do something else when the condition is false? Here comes the C else statement. We can use the else statement with the if statement to execute a block of code when the condition is false. The if-else statement consists of two blocks, one for false expression and one for true expression.

```c
#include <stdio.h>

int main() {
    int age = 10;

    if (age >= 18) {
        printf("Eligible for vote");
    }
    else {
        printf("Not Eligible for vote");
    }
    return 0;
}
```

OUTPUT:

```
"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"
Not Eligible for vote
Process returned 0 (0x0)   execution time : 0.050 s
Press any key to continue.
```

The block of code following the else statement is executed as the condition present in the if statement is false.

3.Nested if-else in C

A nested if in C is an if statement that is the target of another if statement. Nested if statements mean an if statement inside another if statement. Yes, C allow us to nested if statements within if statements, i.e, we can place an if statement inside another if statement.

```c
#include <stdio.h>

int main(){
    int age = 11;

    if (age >= 18) {
        if (age >= 60)
            printf("Eligible to vote (Senior Citizen)\n");
        else
            printf("Eligible for vote\n");
    }
    else {
        printf("Not eligible to vote (Under 18)\n");
        if (age >= 13)
            printf("teenager\n");
        else
            printf("not a teenager\n");
    }

    return 0;
}
```

OUTPUT:

```
"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"
Not eligible to vote (Under 18)
not a teenager

Process returned 0 (0x0)   execution time : 0.046 s
Press any key to continue.
```

4. if-else-if Ladder in C

The if else if statements are used when the user has to decide among multiple options. The C if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the C else-if ladder is bypassed. If none of the conditions is true, then the final else statement will be executed. if-else-if ladder is similar to the switch statement.

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```c
#include <stdio.h>

int main() {
    int i = 20;

    // If else ladder with three conditions
    if (i == 10)
        printf("Not Eligible");
    else if (i == 15)
        printf("wait for three years");
    else if (i == 20)
        printf("You can vote");
    else
        printf("Not a valid age");

    return 0;
}
```

OUTPUT:

"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"

```
You can vote
Process returned 0 (0x0)   execution time : 0.043 s
Press any key to continue.
```

Ultimately, both if-else and nested if-else statements are essential C language tools for directing a program's flow by selecting which code to run based on runtime conditions, with nesting providing the ability to manage increasingly intricate, multi-layered logical decisions.

Swap Two Numbers Without Using Third Variable:

```c
#include <stdio.h>

int main() {
    int a,b;
    scanf("%d %d", &a, &b);

    //Before swap
    printf("a=%d , b=%d\n",a,b);

    a=(a+b)-(b=a);

     //After swap
    printf("a=%d , b=%d",a,b);

    return 0;
}
```

OUTPUT:

```
"C:\Users\Hasd\Downloads\Lab Final\Hasib.exe"
10 20
a=10 , b=20
a=20 , b=10
Process returned 0 (0x0)   execution time : 2.256 s
Press any key to continue.
```